

Entry Delays and Fighting Brands: Evidence from Generics and Authorized Generics*

Rubaiyat Alam

Rena M. Conti

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Abstract

Generic entry – a key driver of lower pharmaceutical prices and improved access – is slowed by regulatory delays and potentially deterred by launch of Authorized Generic (AG) by the brand drug maker. AGs are chemically identical to the brand drug, do not carry the brand label, are priced lower than the brand drug, and can be released at any time. Using data on the US, we develop and estimate a structural model of generic entry and AG release timing to simulate policy counterfactuals. Promoting faster generic entry leads to fewer generic entrants yet lower prices in the early periods of the market. Banning AG leads to more generic entrants but mixed effects on prices – prices are higher in early periods due to missing AG competition, but may be lower in later periods if multiple new generics enter. Therefore, when considering incentivizing or banning post-exclusivity actions by brand drug makers, we should account for generic entry response.

*Corresponding author: Rubaiyat Alam, rubaiyat@ksu.edu. Rubaiyat Alam: Kansas State University. Rena M. Conti: Boston University. The authors thank IQVIA Institute for Human Data Science for providing access to the National Sales Perspectives™ dataset for use in this project. This is a revised version of the third chapter of Rubaiyat Alam's PhD dissertation at Boston University. We are grateful to Marc Rysman for his time and invaluable guidance, without which this project would not have been possible. We thank Mitsuru Igami for detailed comments that improved the paper. We also benefited greatly from extensive discussions with Gianluca Antonicchia, Farasat Bokhari, Randall P. Ellis, Chris Garmon, Jihye Jeon, Albert Ma, and Brian McManus. Finally, we would like to thank Eric Hardy, Chuqing Jin, Ginger Jin, Tim Lee, Thomas McGuire, Brett Wendling, and participants at seminars in Boston University and other institutions for useful comments.

1 Introduction

Generic competition lowers drug prices and improves consumer access, making the economics of generic entry important for policy. Generic entry faces two key hurdles. First, FDA approval is uncertain and lengthy, creating entry delays. Second, the brand incumbent can release an Authorized Generic (AG), a chemically identical but unbranded, lower-priced version of the brand drug. Such “fighting brand strategy” segments the market – the brand serves the high-end brand-loyal segment, and the fighting brand (AG) and generics serve the low-end price-sensitive segment. This additional competition worsens expected profit for generic entrants.¹

The above issues have received attention from policymakers. Reducing regulatory delays to promote faster generic entry has been a priority of the FDA and antitrust authorities since the 2010s. However, the impact of such policies on equilibrium market outcomes is not well understood. Similarly, debates about whether AGs are procompetitive or anti-competitive led to a report by the Federal Trade Commission ([FTC \(2011\)](#)), have surfaced in Europe ([Appelt \(2015\)](#)), are summarized in [Feldman \(2021\)](#), and have received media coverage ([Appendix G](#)). The fundamental issue is that while AGs provide additional price competition, they may also deter generic entry which reduces price competition. The key uncertainty is to what extent AG deters generic entry, which our paper aims to address.

We use structural estimation to simulate how regulatory delays and AG releases affect generic entry decisions and market outcomes. The supply side of the industry is modeled in two stages. First, generic makers decide whether to apply for entry into a market. Second, a dynamic game begins where every period, generics who applied to enter receive approval stochastically, the brand drug maker decides whether to release an AG, and market participants compete in prices. Regulatory delays manifest as generics receiving FDA approval stochastically over time after loss-of-exclusivity (LOE) rather than being able to launch immediately upon LOE. This two-stage model setup reflects the institutional reality that unlike AGs, generic entry faces uncertain regulatory delays, and thus generics need to apply for entry in advance. We solve the two-stage supply model by backward induction, so AG and generics form rational expectations about each other’s entry and pricing decisions

¹Such fighting brand strategies are also commonly observed in other industries. An example (from [Bourreau et al. \(2021\)](#)) in the microprocessor industry is Intel with the Pentium series (brand) and Celeron (fighter brand) to compete with AMD.

when making a choice.

We use data on national sales and revenue for prescription drugs in the US between 2004-2016 (IQVIA National Sales Perspective) to estimate the structural parameters. We define “loss-of-exclusivity” (LOE) as the first quarter when any generic enters a molecule-formulation. We measure regulatory delays as the time between LOE and generic entry, to study how quickly generics enter once exclusivity expires. We use average entry rates by quarter-since-LOE to capture this, which generics use to form expectations about entry dates in our model. Prices in different industry states are predicted using a reduced-form regression to avoid making pricing conduct assumptions.

For our counterfactuals, we solve the model for each market. An important element of industry dynamics is that the early quarters after LOE are most profitable for generics and AGs – few generics are present as the rest wait for FDA approval, resulting in lower price competition and higher market shares.

Our first set of counterfactuals illustrates how entry deterrence operates within our model. Both generics and AGs can deter each other’s entry in different ways. Generics can deter AG because they move first, i.e. they apply for entry and incur the entry cost first. However, AGs can also deter generics because they can launch their product earlier than generics – the early launch significantly reduces generics’ expected profits by providing price competition in the early profitable periods after LOE. To showcase these two forces, we first show that no AG is released when the generics can launch immediately and the AG has no cost or demand advantage over independent generics; we then show that AG deters more generics when it can launch earlier rather than being forced to delay.

Second, we study the impact of reducing regulatory delays to promote faster generic entry. Reduced regulatory delays lead to weakly fewer generic entrants yet lower prices in the early periods of the market. This is because while reduced delays mean each generic operates for more periods, its rivals also launch earlier in the market. Each generic faces intense price competition as soon as it enters, and this more than offsets the advantage from operating for more periods. Lifetime generic profits fall, and thus fewer generics choose to enter.² Despite fewer generic entrants, prices are lower in early periods of the market.

²Another way of seeing this is that increasing regulatory delays creates an incentive for an additional generic entrant to take a gamble and apply for entry – while it makes the same per-period profit in the long run with or without regulatory delays, there is now a chance that its rivals will be slow to obtain approval while it gets early approval. This leads to higher profits for the generic in the short run compared to the case

This is because while there are fewer generics, they all launch almost immediately, again leading to intense price competition from the beginning. However, fewer generics may result in higher prices in later periods.

Third, we study the effect of an AG ban on short-term and long-term market outcomes. We show that a ban on AG generally leads to an additional generic entrant and sometimes multiple new entrants. The reason an AG may deter multiple generics is because the AG has an early-launch advantage – it often chooses to launch during the early profitable periods of the market, and so has a larger negative impact on rivals’ expected lifetime profits compared to an independent generic which is slowed by regulatory delays. The effect of AG ban on prices is mixed. When an AG is released, it is usually in the early periods of the post-LOE market. Therefore, with the AG ban, generics entering the market early do not face an additional competitor through the AG; thus, prices are higher in the early periods of the market. However, if the AG were deterring multiple generics from entering, then prices in the later periods of the market would be lower with the AG ban as multiple new generics choose to enter.

Overall, our paper argues that any assessment of incentivizing or banning post-LOE actions by the incumbent should account for generic entry response. Within our setting, policies encouraging or banning AGs are offset by generic response, attenuating the impact on consumer welfare.

Contributions and related literature: This paper contributes to the following strands of literature. First, we contribute to a small literature on Authorized Generics. We are the first to study the AG release decision and its impact on generic firms by using a structural model of entry. [Appelt \(2015\)](#) uses a recursive bivariate probit regression to argue that AG entry does not impact generic entry in Germany; in contrast, our paper uses data on prices and market shares combined with a model of industry dynamics to study this question for the US. A report from the Federal Trade Commission ([FTC \(2011\)](#)) mainly focuses on how AG release may disincentivize Paragraph IV lawsuits by generics and does so by compiling industry data and company documents. Our descriptive findings broadly mirror those of [Fowler et al. \(2023\)](#), who use industry data to describe broad trends about AGs over 2010-2019; see Appendix F for a comparison. [Berndt et al. \(2007\)](#) presents descriptive statistics and case-studies on AGs, and uses them to speculate about the effect of AGs on market

with no regulatory delays.

outcomes; our paper formalizes and simulates these economic effects.

Second, we construct a simple structural model of generic industry and pharmaceutical product entry that captures generic industry dynamics, yet is quick to solve and can be easily extended in various directions. While our structural model is strictly about AGs, we believe it can be extended to study other strategic reactions against generics by brand drug makers (product line extensions and reformulations, additional end-of-life indications, etc.) and other regulatory delays (such as those faced by biosimilars). In particular, the model can be used to tractably study generic entry deterrence. In doing so, we add to the literature on generic drug entry and deterrence. In contrast to other papers, we develop a parsimonious model that captures generic industry dynamics yet is easy to solve and extend. Other notable papers studying generic entry using structural models are [Ching \(2010\)](#), [Gallant et al. \(2017\)](#), [Scott Morton \(1999\)](#), [Reiffen and Ward \(2005\)](#), [Starc and Wollmann \(2023\)](#), and [Wang et al. \(2022\)](#); see [Ching et al. \(2019\)](#) for an overview. [Wang et al. \(2022\)](#) uses a static entry model and reduced-form profit functions to understand factors (particularly firm-specific heterogeneity that we ignore) that promote generic entry into a molecule-formulation; in contrast to our paper, they do not explicitly account for dynamics or AGs, and do not track market shares or prices. As a result, we are better positioned to study the impact of dynamic policy changes such as faster entry rates, which they also study and reach the opposite conclusion to ours. More generally, our paper relates to a large literature on generic competition and pharmaceutical pricing; key references for our paper include [Arcidiacono et al. \(2013\)](#), [Berndt et al. \(2007\)](#), [Bokhari and Fournier \(2013\)](#), [Chen et al. \(2022\)](#), [Dubois et al. \(2022\)](#), [Ellison and Ellison \(2011\)](#), [Frank and Salkever \(1992\)](#), [Frank and Salkever \(1997\)](#), [Ganapati and McKibbin \(2023\)](#), [Hermosilla and Ching \(2023\)](#), [Lakdawalla \(2018\)](#), [Olson and Wendling \(2018\)](#), [Reiffen and Ward \(2007\)](#), and [Tenn and Wendling \(2014\)](#); see [Scott Morton and Kyle \(2011\)](#) for an overview. [Berndt et al. \(2017\)](#), [Berndt et al. \(2018\)](#), [Conti and Berndt \(2020\)](#), and [Frank et al. \(2021\)](#) provide important descriptive evidence about the aggregate dynamics and inner workings of the generic industry which we use to motivate our model.

Third, we contribute to a somewhat limited empirical literature on entry deterrence and preemption. The fundamental empirical problem is that when a potential entrant does not enter, the researcher cannot tell if this was due to deterrence or due to absence of entry threat to begin with (see [Cookson \(2017\)](#) and [Gil et al. \(2024\)](#) for a detailed discussion

and creative workarounds). In our paper, we tackle this by building a model of per-period profits and entry choices, and using it to predict entry decisions in different markets. The sequential nature of moves in this institutional setting (i.e., generics make entry decision first followed by AG) leads to a clean solution of the entry deterrence problem. Empirical papers on this topic include [Cookson \(2017\)](#), [Fang and Yang \(2024\)](#), [Gil et al. \(2024\)](#), [Goolsbee and Syverson \(2008\)](#), [Igami and Yang \(2016\)](#), and [Schmidt-Dengler \(2024\)](#). There is extensive theoretical literature on this topic, which can be found in the references above.

Fourth, we contribute to a sparse Empirical IO literature on fighting brands. We are one of the few papers to build and estimate a structural model of an incumbent releasing a fighting brand. Fighting brands are widely observed in many industries besides pharmaceuticals, so our model and results add to our understanding of such strategies, particularly entry deterrence motives. [Bourreau et al. \(2021\)](#) study how fighting-brand launches marked a breakdown of collusion in French mobile telecommunications. There is a theoretical literature on fighting brands, notably [Anderson and Dana \(2009\)](#), [Deneckere and McAfee \(1996\)](#), and [Johnson and Myatt \(2003\)](#).

Finally, our model combines elements from the Empirical Industrial Organization literature on static entry games and dynamic games. For static entry, key references include [Berry \(1992\)](#), [Berry et al. \(2016\)](#), and [Seim \(2006\)](#). For dynamics, some key references for our setting are [Gowrisankaran and Rysman \(2012\)](#), [Igami \(2017\)](#), [Igami \(2018\)](#), [Igami and Uetake \(2020\)](#), and [Rust \(1987\)](#); see [Aguirregabiria et al. \(2021\)](#) for a detailed overview of the literature.

2 Institutional Background

To enter a market after the brand drug maker loses exclusivity, generics must apply for FDA approval. However, gaining FDA approval is lengthy, costly, and takes uncertain time. The brand drug maker can respond to generic entry by releasing an Authorized Generic (AG). The AG is chemically identical to the brand drug but without the brand label attached. Unlike generics, the AG can be launched at any time without having to wait for FDA approval. The general patterns after loss of exclusivity are: brand drug prices remain high, generics and AG charge low price which further declines with more competition, and the brand drug loses most of its market share to generics and AGs over time.

2.1 Generic Drugs and Market Entry

A pharmaceutical maker invests in R&D to discover a drug, then patents the molecule and conducts tests for efficacy and safety (see [Dubois et al. \(2015\)](#) and [Khmelnitskaya \(2023\)](#), and the references therein). This drug is commonly referred to as the “brand drug”.

After the brand drug’s loss-of-exclusivity (LOE), generic drugs can enter the molecule-formulation once they receive final approval from the FDA. The loss-of-exclusivity may be brought about by patent expiration, patent challenges, or settlements.

The generic maker submits an Abbreviated New Drug Application (ANDA) to the FDA for approval. To gain FDA approval, generic makers must prove to the FDA that they can manufacture products that are “bioequivalent” to the brand drug, at scale for commercial distribution, and according to Current Good Manufacturing Practices.³ The cost incurred to receive FDA approval ranges from \$3 million to \$15 million.⁴

The FDA’s approval process is lengthy, of uncertain duration, and sometimes slowed down by litigation issues. The mean approval time for an ANDA is between 32-40 months for our data period, so generics typically submit their ANDAs well before LOE.

In this paper, we refer to applying for FDA approval as “applying for entry”, and refer to generics launching their product after FDA approval as “entry”.

2.2 Authorized Generics

In response to generic entry, the brand drug maker can release an Authorized Generic (AG). The AG is chemically identical to the brand drug in terms of molecule structure and formulation, except it does not have the brand label attached. The AG competes directly with

³Bioequivalence means “product has the same active ingredient, dosage form, strength, route of administration and conditions of use” as the reference brand drug. An important aspect of proving bioequivalence is that the generic firm needs to reverse-engineer the brand drug. Since some aspects of the manufacturing process for the brand drug may be undocumented or kept as trade secrets by the brand maker, generic firms must experiment with production to meet regulatory requirements. Proving the generic drug is bioequivalent to the brand and can be made at scale according to good manufacturing practices typically requires significant effort and may entail multiple rounds of back-and-forth discussions with the applicant until the FDA is satisfied.

⁴A range of generic entry cost estimates can be found in [Berndt et al. \(2017\)](#), [Gottlieb \(2016\)](#), [Scott Morton \(1999\)](#), and [Starc and Wollmann \(2023\)](#), among others. The range we state is consistent with these estimates.

the independent generics in appearance and prices.⁵

A key detail about product launches in this setting is that AGs can be introduced any-time, unlike generics. This is because they are riding on the brand drug's approval from the FDA, and therefore do not require additional approval.

Why would a brand drug maker not release an AG every time it faces generic entry? There are several reasons. First, the AG would cannibalize the brand drug's sales. Second, the AG entry would lead to generics lowering their prices in response, further increasing the price gap between brand and generics; this may cause brand drug sales to fall even more than without AG entry. Third, it could be that many generics enter in the very first period after loss-of-exclusivity, meaning that the AG will make little profit due to intense price competition; this explains why AGs have a strong incentive to enter during the 180-day exclusivity period, as detailed in [FTC \(2011\)](#). Fourth, there could be policy uncertainty such as how the FDA would approve or regulate the new generic entrants, or how a lawsuit might be resolved, that may add further uncertainty to the release decision. Fifth, it could be due to supply-side reasons such as manufacturing delays, organizational frictions, inexperience with generic production,⁶ and legal settlements.⁷

The practice of AG release has generated considerable controversy. It led to the FTC

⁵Patients and physicians cannot easily tell whether a generic product is an independent generic or an AG, or whether an AG is available for a molecule-formulation. This is by design, since the AG is meant to compete with generics without cannibalizing the brand drug's sales.

⁶The brand drug maker may not have any generic subsidiaries, which makes it very difficult to release an AG, because marketing a nonbrand drug versus a brand drug entail very different skillsets. The marketing of a brand drug focuses on advertising and detailing, negotiating formulary positions by insurer, etc. The marketing of a generic drug focuses on extreme cost efficiency, winning pharmacy auctions ([Cuddy \(2020\)](#)), etc. These are very different skillsets, and for a brand drug, it may not be worth the hassle of getting into the generics business just to release an authorized generic.

A brand maker also does not always directly release its own AG. Instead, it may be done by a generic subsidiary of the brand, or through contracting with another generic firm, or by firms specializing in releasing AGs on behalf of other makers (such as Prasco). In these cases, the nonbrand firm takes over part of the responsibilities and gets a share of the profits. See Appendix [H](#) for more details from [FTC \(2011\)](#).

⁷This raises the question of why the brand maker does not always match the price of generics with the brand drug instead of releasing the AG. First, there is a price discrimination motive where consumers who like the brand drug are willing to pay a high price for it, so having the expensive brand and inexpensive AG in the market allows the maker to capture both types of consumers. Second, if the generic entrant has to leave the market (e.g. due to legal reasons, patenting issues, technical or manufacturing problems) the brand may not be able to raise its prices back up without facing backlash from buyers. Third, if the brand product's price is lowered, it is hard to justify high prices on line extensions. All the points mentioned above are corroborated in [FTC \(2011\)](#).

releasing a report in 2011 ([FTC \(2011\)](#)) studying the impact of AGs on Paragraph IV lawsuits, prices, and generic revenues. The report concluded that while AGs reduce generic profits, they have a limited impact on generics filing Paragraph IV lawsuits. They reach their conclusion using a mix of internal documents and break-even analysis. Much of the controversy about AGs is summarized in [Feldman \(2021\)](#). The issue has also received media coverage; see Appendix [G](#).

The debate on AGs can be summarized as follows. AGs may provide additional price competition and lower prices even further. However, the presence of AGs may deter generic entry, which will reduce price competition. The effect on generic entry is the key uncertainty in these articles, and this is what our paper aims to shed light on.

2.3 Delays in Generic Entry

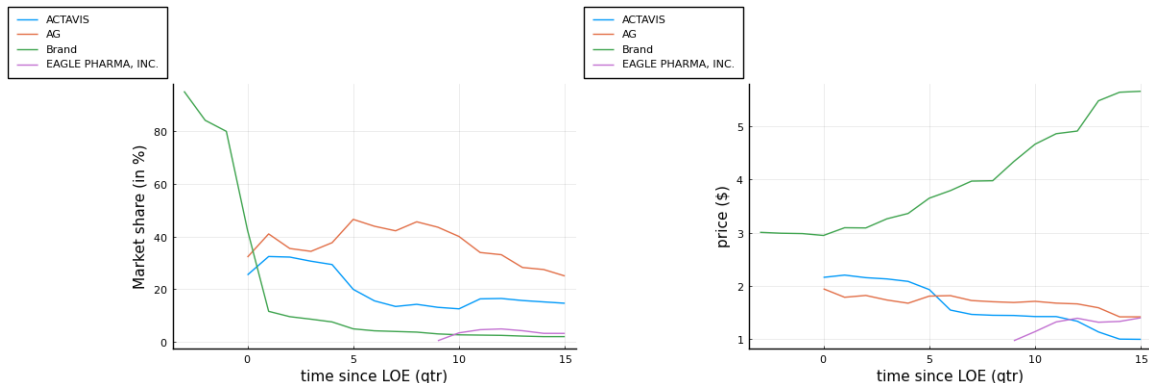
Generics face various regulatory delays that slow down entry into a molecule-formulation. Some examples are as follows. First, the most important is the time FDA takes to assess ANDA applications and give approval, which often involves a back-and-forth between the FDA and the maker regarding the production process. Over our dataset’s period, the FDA suffered from large backlogs of ANDA applications waiting to be approved, although this has improved.⁸ Second, the Hatch-Waxman Act set up a 180-day first-filer-exclusivity provision where the first generic to file for Paragraph IV certification enjoys a 180-day period after entry when other generics cannot enter; thus, a rival becoming the first-filer stops other generics from entering the market for 180 days.⁹ Third, FDA approval of ANDAs may be held up by litigation issues, which commonly arise when a brand’s patent is being challenged by generics.¹⁰ Finally, there may also be information issues due to legal or technical reasons such that the generic maker cannot correctly anticipate the date

⁸Final approval by the FDA requires inspections to meet CGMP (Current Good Manufacturing Practice Regulations) at scale, and they can fail. Furthermore, for certain types of molecules and formulations, the generic maker can face manufacturing problems when setting up production lines for FDA approval, delaying approval further. Unrelated inspections for the first-filer can fail, which then cascades to interrupting final approvals for all subsequent filers. There have also been reports of brand drug makers delaying providing samples for generic makers to copy in an effort to slow down ANDA approvals.

⁹For a brief overview of these topics, see Appendix [F.5](#).

¹⁰There are a variety of ways in which the brand manufacturer can slow down litigation and therefore FDA approval. In particular, settlements with the first filer generic that delay their launch may hold up approval of later generics, an issue known as the “approval bottleneck”.

Figure 1: Evolution of market shares (left) and prices (right) for Diclofenac-Misoprostol (oral formulation).



of LOE, leading to delays in filing and hence approval.

For a generic in a molecule-formulation, the “delay” we study is the gap between loss-of-exclusivity (LOE) date and the generic’s launch date. Here LOE is defined as the first generic’s launch date in that molecule-formulation. Our measure of delay is what essentially matters in our setting – if there is an exclusivity protecting the brand drug, then no generic is approved until the exclusivity disappears. Thus the question is, once exclusivity expires, how quickly do generics enter and drive down prices. Our measure of delay is therefore different from the gap between a generic’s ANDA filing date and its launch date.

2.4 Evolution of Market Shares and Prices

The general pattern we see in the data regarding market prices and shares is as follows. After loss-of-exclusivity by the brand drug, generics enter the market and undercut the brand’s price, and the brand drug’s price remains above that of generics. A brand drug maker sometimes releases an AG in response to generic entry. The AG and generics set prices close to each other. Over time, the brand drug’s market share falls dramatically, while those of nonbrand drugs rise. An example of this pattern is presented in Figure 1.

3 Data and Descriptive Statistics

We leverage data on quarterly sales and revenue of prescription drugs in the US from 2004 to 2016, and use this to construct market shares and prices. We define a market as a molecule-formulation; of the 240 molecule-formulations we restrict our attention to, about 45% see an AG released. Most AGs are released soon after the first generic entry occurs. Most markets have between 1-11 competitors. We run descriptive regressions to show how market share and prices vary in our dataset and corroborate the patterns stated in the previous section.

3.1 Data

The data for this paper comes from IQVIA’s National Sales Perspective™(NSP) database and covers sales in the US for 2004 - 2016. This dataset contains national sales of prescription drugs to pharmacies, clinics, hospitals, and other distribution channels, aggregated to the quarterly level. Each observation includes the name of the molecule (active ingredient) and brand name (if relevant), quarter, sales amount in drug unit volume, sales amount in US dollars, supplier of the product, distribution channel, product form (e.g., oral, injectable), and other information including therapeutic class, product launch date, etc.

We define a “drug” to be a group of products having the same molecule (or active ingredient) and product form, following previous literature. A drug’s “therapeutic class” describes what broad diagnoses it targets; a drug’s “molecule” describes the active ingredients present; and a drug’s “formulation” is its delivery method, e.g. oral, injectable, etc.

We now give a brief overview of the dataset; this closely follows [Berndt et al. \(2017\)](#), which contains a more detailed description of the dataset. It is constructed through a projected audit that tracks drug sales from makers, wholesalers, and distribution centers that serve national and regional chain pharmacies, grocery stores, etc. to pharmacies and other distribution outlets. The measure of “revenue” for each drug is the total amount paid for product purchase to the drug maker. Furthermore, the dataset reports volumes of each drug sold in “standard units”, where a standard unit is defined as the smallest unit of the drug standardized across dosage and therapeutic form ([Dubois and Lasio \(2018\)](#)). We compute prices as the ratio of total revenue to total quantity in standard units, following [Dubois et al. \(2022\)](#) among others.

Data on Authorized Generics were hand-collected.¹¹ AGs were identified from the FDA’s maintained list of AGs, an online source maintained by some producers of AGs, and verified with news reports on AG releases. See Appendix F.1 for links to data sources.

We define the “loss-of-exclusivity” (LOE) period for a molecule-formulation as the first quarter when any generic appears in that molecule-formulation. This suffices for our paper as we are interested in the periods following observed generic entry into a market. The date of the first entrant may depend not just on the exact date of patent expiration, but also patent challenges and settlements.

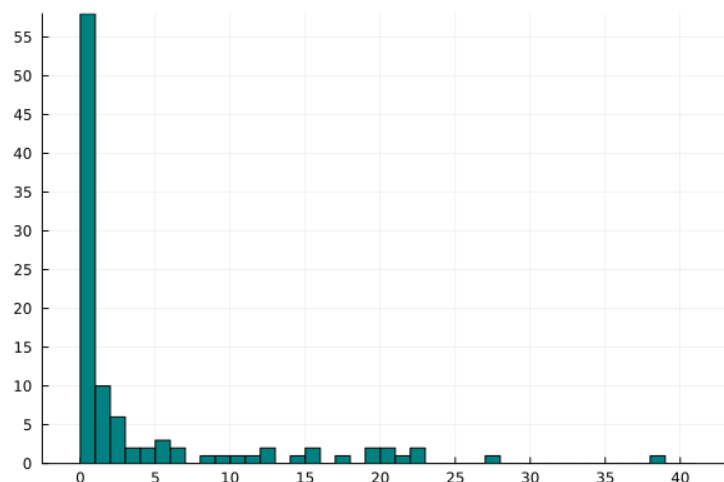
To summarize, for each drug product we see quarterly sales in the US, revenue generated, list of active ingredients, formulation of the drug (oral, injectable, or other), and therapeutic class (ATC3). Sales for a product-molecule-formulation are aggregated by dosage and strength. Finally, we define a market at the molecule-formulation level, following previous papers in the literature (Conti and Berndt (2020), Frank et al. (2021), Wang et al. (2022), Yurukoglu et al. (2017)).¹²

¹¹We searched news stories about every molecule online to try to ensure there were no non-economic events surrounding them that would affect generic entry. Molecules that faced unnatural events, such as lawsuits being settled then revived, facing new lawsuits after entry and being forced to discontinue, facing public pressure (such as EpiPen), etc. are left out of the final sample.

¹²This raises the question of why we define a market at the molecule-formulation level instead of a higher level, such as the therapeutic class. First, we include a molecule-formulation fixed effect in the demand function to account for other molecule-formulations in the same therapeutic class. Second, similar market definitions have been used in various antitrust proceedings and empirical papers (Conti and Berndt (2020), Frank et al. (2021), Starc and Wollmann (2023), Wang et al. (2022), Yurukoglu et al. (2017)). Third, this allows us to capture substitution patterns between bioequivalent products more accurately. Fourth, the brand drug’s patent and generic approval holds at the molecule-formulation level. Fifth, one drug can be in multiple therapeutic classes, making modeling therapeutic classes as markets unclear. Finally, evidence suggests that while there can be substitution across molecule-formulations within the same therapeutic class, the substitution is usually from a molecule-formulation without a generic to a molecule-formulation with one (Arcidiacono et al. (2013)).

The fact that different formulations are protected by separate patents bears particular emphasis. The literature suggests that different formulations are usually protected by separate patents, i.e. oral and injectable versions have different patents. There is an extensive literature on such “secondary patents”; for example, see Gupta (2024) and Kapczynski et al. (2012). This is both because additional formulations are sometimes considered significant innovations, and because of strategic patenting by brand manufacturers to extend exclusivity. This then suggests separate 180-day exclusivity periods for separate formulations. All these considerations imply generic entry decisions are likely made at the molecule-formulation level.

Figure 2: Histogram of time-difference between first generic entry and AG release period.



Notes: This figure is limited to 22 quarters, while in the dataset 4 AGs get released later than 22 quarters. This distribution has a 10th percentile of 0, 25th percentile of 0, median of 0, 75th percentile of 4.25, and 90th percentile of 16.4. See Appendix D for the exact distribution.

3.2 Descriptive Statistics

While our dataset has comprehensive coverage across all prescription drugs sold in the US, we look at a smaller subset of drugs for our project. We select molecule-formulations that saw its first generic entrant after 2004 and before 2016, which are not biologics, for which we can identify presence or absence of AG with certainty, and those that are not surrounded by strong external circumstances (e.g. media/public outrage, serious or repeated lawsuits). This leaves us with 240 molecule-formulations, listed in Appendix I.¹³

¹³We now discuss how this sample selection ties in with our research question and interpretation of results. For studying AG entry, we believe this way of identifying markets AG may enter is quite realistic. AGs are always only released in response to generic entry. In fact, FTC (2011) shows company documents where, when outsourcing AG manufacturing, the brand maker imposes strong restrictions on the partner AG makers to prevent them from launching the AG before the first generic has entered. Therefore, identifying markets with at least one generic as markets where the AG may enter is realistic, and so our AG cost estimation should be unaffected.

One type of market we rule out with this is a market where no generic enters, but would have entered had it not been for the threat of AG release; see Appendix A.4 for a brief discussion. We also rule out markets where no generic enters because of settlements, profitability concerns, or litigation issues. Most high-revenue, clinically important, small-molecule brand prescription drugs do eventually face generic entry.

Table 1 shows descriptive statistics about our selected molecule-formulations: of the 240 molecule-formulations, about 45% see an AG released, and over 67% are oral formulations. Note that each molecule-formulation has one brand drug and can have at most one AG. See Appendix F.2 for a list of brand drug makers releasing AG.

Table 1: Descriptive statistics

	No. of molecule-formulations
Total	240
Formulation: Oral	162
Formulation: Injectable	58
Formulation: All Others	20
AG released	108
AG released in Non-Injectables	99

To illustrate AG release timing, Figure 2 shows presents a histogram of how many quarters after first-generic-entry is AG released. 58 markets see AG release immediately with the first generic entry, 10 markets see AG released one quarter after the first generic entry, and 6 markets see AG released two quarters later. AGs continue being released at lower rates after the third quarter.

We look at the competitive landscape in our selected markets, specifically regarding generic presence. Most markets have 1-11 generic competitors, with the maximum number of generics in a molecule-formulation being 19 in our sample. See Figure 5 in the Appendix for the distribution of total generics that enter into a molecule-formulation.¹⁴

Next we study how market shares and prices vary based on brand labels and levels of competition. To do so we regress market shares and prices on a variety of covariates; the results for non-injectables are reported in Table 2. There are several notable patterns here. More non-brand products lead to greater splitting of the market share and lower prices, but the effect diminishes with the number of nonbrands. The brand has an advantage in terms of attracting market share and also charges higher prices. The brand’s price stays the same or rises as more generics come in – this pattern has been documented previously in the

¹⁴We check how often brand drug exits the market after LOE. Of our 240 molecule-formulations, 31 see the brand drug exiting. For these 31 markets, three see an AG released. Given how infrequently the brand drug exits the market, and the computational burden it will add to our estimation, we do not model brand drug exit decisions.

Table 2: Regressions on market shares and prices for non-injectable molecule-formulations.

	log(market share)	log(price)
	(1)	(2)
Count of non-brands	-0.210*** (0.009)	-0.297*** (0.008)
Count of non-brands squared	0.009*** (0.001)	0.007*** (0.000)
Brand	0.623*** (0.035)	0.104*** (0.026)
Brand x qtrs since LOE	-0.044*** (0.002)	0.007*** (0.002)
Authorized generic	0.168*** (0.023)	0.245*** (0.020)
Brand × Count of non-brands	-0.172*** (0.013)	
Brand × Count of non-brands squared	0.012*** (0.001)	
Brand × Count of generics		0.256*** (0.004)
Brand × AG presence		0.195*** (0.029)
Molecule-Formulation Fixed Effects	Yes	Yes
Obs.	14,352	14,352
R-squared	0.389	0.883

Notes: The variable “qtrs since LOE” is the number of quarters since loss-of-exclusivity. “AG presence” is an indicator for whether an authorized generic is present in the market. “Generic” and “Brand” are indicators for generic and brand drugs respectively.

literature ([Frank and Salkever \(1997\)](#) and subsequently others). The brand also appears to lose market share as more time goes by.¹⁵ The AG has a small advantage over independent

¹⁵The time trend potentially captures the following issues. Brand drug makers can sometimes take actions that slow down generic adoption. This involves actions such as product hopping, getting late-stage-lifecycle approvals for new indications, etc. These actions hinder consumers and pharmacists from switching to generic drugs and getting around automatic substitution laws. While our model does not explicitly account for such actions, our demand specification contains a brand-specific time trend that captures the idea that the brand drug loses its appeal over time as the maker runs out of such tactics. As expected, we find that the brand drug

generics when it comes to market share, and also charges slightly higher prices; we validate this in Table 14. See Table 15 for similar results on injectables. Finally, Table 12 in the Appendix has additional descriptive statistics on market shares and revenues.

Finally we look at generic entry timing in our dataset. Recall that our measure of delay is the time between LOE and a generic’s entry — capturing how quickly competition emerges once exclusivity expires. We define the “fraction of generic entries” at any quarter-since-LOE as the number of generic entries in that timeframe in our sample, divided by the total number of generic entries observed in our sample.

Figure 3 shows the fraction of generic entries happening by quarters-since-LOE. The most notable pattern is that a large fraction of the generic entry happens in the quarter when LOE happens (over 30% of all generic entries). Since generics usually apply for FDA approval before LOE; this suggests that a fraction of generics receive approval to launch by the LOE period. However, the majority of generic entry happens after the first quarter. By 20 quarters-since-LOE, about 93% of the generic entries have taken place. The spike around 3 quarters-since-LOE reflects some generics entering after waiting out the 180-day exclusivity period of a rival generic.

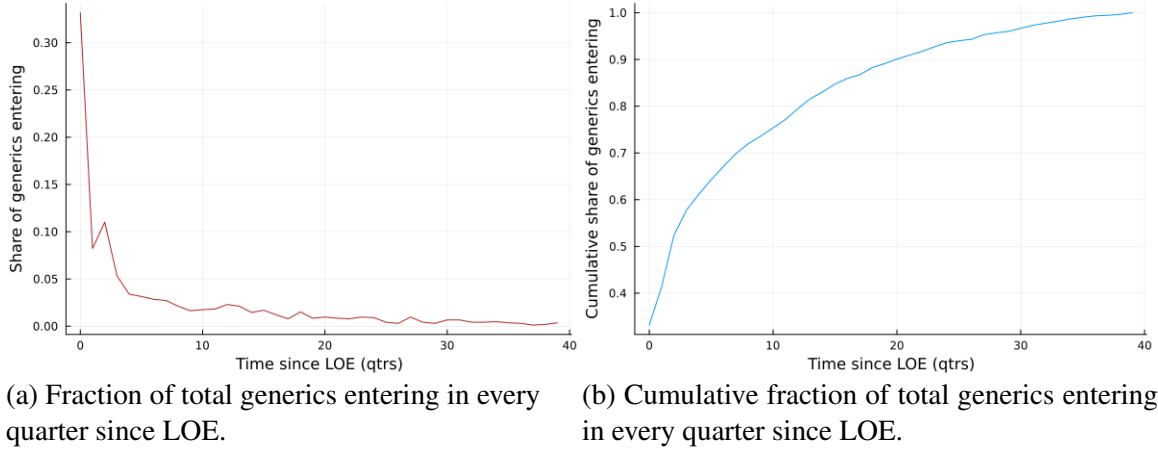
We also confirm several statements using ANDA filing and approval dates from [Feldman et al. \(2016\)](#). That dataset collects filing dates and approval dates of all ANDAs for a subset of molecules and formulations. Appendix J shows some descriptive statistics from this data. We find that ANDA approval dates within a molecule-formulation can be significantly far apart, from several months to years. The time between ANDA filing and approval usually takes around several years, with a median of 35.3 months. Importantly, we recreate Figure 3 with ANDA approval dates. That is, we calculate the delay between first ANDA approval and subsequent ANDA approvals within the same molecule-formulation. The result, shown in Figure 7, is very close to our data despite using approval dates rather than launch dates. This also helps rule out an alternative model where generics all receive final approval as soon as LOE occurs, and the variation in entry times is purely manufacturing frictions that stagger out product entries over time.¹⁶

becomes less attractive over time, possibly because such tactics become less effective over time.

¹⁶Such a model would have been pretty unlikely to begin with, as generics generally launch as soon as they receive approval because they want to start recouping their sunk cost of entry, and the earlier periods of the market are the most profitable.

[Wang et al. \(2022\)](#) uses exact dates from the Orange Book and also recover a figure similar to Figure 3.

Figure 3: Generic entry rate.



4 Model

To study the interaction of generic entry and AG release decisions, we develop a structural model of entry and price competition between AG and generics. The aim is to develop a simple tractable model that captures the important economic mechanisms. We construct a two-stage model of pharmaceutical product entry. The first stage is a static game of generic entry. The second stage is a dynamic game of Authorized Generic release decision. Every period, generics who applied to enter are stochastically approved for entry by the FDA, the brand drug maker decides whether to release an AG, and the market participants compete on prices. Section 4.3 goes over the reasoning behind the modeling choices.

4.1 Overview

We first present an overview of our supply model. Generic makers are slowed down from entering due to regulatory delays; as a result, they apply for entry well in advance of LOE. In contrast, AGs can enter at any time. This motivates our model setup below.

There are two stages to the supply side:

1. Stage 1: Generic firms decide whether to apply to enter a market or not. This stage is modeled as a game of simultaneous entry application between identical firms. The

generics are forward-looking and form rational expectations about AG entry and regulatory delays when making entry decisions.

2. Stage 2: Loss-of-exclusivity happens and a dynamic game begins. The dynamic game lasts T periods. Every period, a stochastic number of the generic firms which applied for entry are approved and enter into the market. The brand drug maker decides whether to release AG or not (where the AG release decision is irreversible). Then, all the firms set prices and receive payoffs.

4.2 Model setup

We now explain the model in reverse, starting from payoffs in the second stage. Throughout we suppress the vector of structural parameters from function arguments for notational clarity.

4.2.1 Second-stage

Consider firms operating in molecule-formulation m .

The brand drug maker's per-period payoff is:

$$\pi^b(s_{mt}) = [P_{mt}^b - MC_m^b]s_b(s_{mt})M_m + \mathbf{1}(AG_{mt} = 1) \left[[P_{mt}^{AG} - MC_m^{AG}]s_{AG}(s_{mt})M_m \right] \quad (1)$$

where s_{mt} denotes the state variable vector in market m , superscripts and subscripts b and AG refer to the brand product and AG respectively, P denotes price, MC denotes marginal cost, $s(s_{mt})$ denotes market share, and M_m denotes market size. The indicator variable $\mathbf{1}(AG_{mt} = 1)$ denotes whether an AG has been released into the market. The intuition behind the brand drug maker's payoff is that it makes the first part of the payoff from the brand drug, and the second part from the AG conditional on releasing the AG.

Similarly, the generic firm l 's per-period payoff with the firm-specific state variable vector $s_{l,t}$ is:

$$\pi^g(s_{lmt}) = (P_{mt}^g - MC_m^g)s_g(s_{lmt})M_m \quad (2)$$

The brand drug, generics and AG compete on prices. We use a reduced-form price prediction regression to predict prices in different states and markets.

A dynamic game begins from $t = 1$ and lasts T periods, where every period is a quarter. In period $t = 2$ the brand drug’s market exclusivity expires, and every period a random number of generic firms gain FDA approval and enter the market. Thus, t is a measure of periods relative to LOE of a molecule-formulation, rather than measuring calendar time.

Since AG release is a dynamic decision, the brand drug maker’s choice problem can be written in terms of value functions. Thus, the value function for a brand drug maker every period is given by:

$$V^b(s_{mt}, \epsilon_{mt}) = \max_{AG_{mt+1} \in \{0,1\}} \pi^b(s_{mt}) - \mathbf{1}(AG_{mt} = 0, AG_{mt+1} = 1) \kappa_m^{AG} + \beta E[V^b(s_{mt+1}, \epsilon_{mt+1}) | s_{mt}, \epsilon_{mt}] + \epsilon_{mt}(AG_{mt+1}) \quad (3)$$

where $AG_{mt+1} = 1$ means the AG is in the market m during period $t + 1$. AG entry cost in molecule-formulation m is denoted as κ_m^{AG} . The AG entry decision is irreversible, i.e. $AG_{mt} = 1$ implies $AG_{mt+k} = 1 \forall k > 0$. AG entry is implemented with a one-period delay.

The timing of AG release could be influenced by factors observed to the brand drug maker but unobserved to the econometrician. To account for such factors, the model includes choice-specific payoff shocks $\epsilon_{mt}(AG_{mt+1})$. These shocks follow a Type 1 Extreme Value distribution, and have a logit scaling parameter σ_ϵ that governs the variance of these shocks.

The probability of releasing an AG at time t conditional on not having been released by time $t - 1$ is¹⁷:

$$Pr(AG_{mt+1} = 1 | s_{mt}, AG_{mt} = 0) = \exp\left(\frac{-\kappa_m^{AG} + E[V^b(s_{mt+1}, AG_{mt} = 1 | s_{mt}, AG_{mt} = 0)]}{\sigma_\epsilon}\right) / B \quad (4)$$

¹⁷This expression involves a slight abuse of notation: s_{mt} already includes the value of AG_{mt} , but we separate them out in the arguments here for clarity.

where

$$B = \exp\left(\frac{-\kappa_m^{AG} + E[V^b(s_{mt+1}, AG_{mt+1} = 1 | s_{mt}, AG_{mt} = 0)]}{\sigma_\varepsilon}\right) + \exp\left(\frac{E[V^b(s_{mt+1}, AG_{mt+1} = 0 | s_{mt}, AG_{mt} = 0)]}{\sigma_\varepsilon}\right)$$

The probability of releasing an AG conditional on having released AG in the past is one, i.e. AG release is an irreversible decision:

$$Pr(AG_{mt+1} = 1 | s_{mt}, AG_{mt} = 1) = 1 \quad (5)$$

The value function for generic l is given by:

$$V^g(s_{lmt}) = \pi^g(s_{lmt}) + \beta E[V^g(s_{lmt+1}) | s_{lmt}] \quad (6)$$

where s_{lmt} includes whether generic l has been approved for production by the FDA in market m . While the generic makers do not face a dynamic decision in the second stage, they use the value function to make entry decision in the first stage.

From period T , all drug makers make zero profits, i.e. $\pi^b(s_{mt}) = \pi^g(s_{lmt}) = 0$ for $t \geq T$. This accounts for small economic profits in mature drug markets due to new drug discovery, patent expiration of other brand drugs, and the extremely-low profit margins generally observed in a mature generic drug market.

The drug makers have to form expectations over generics' entry probabilities; recall that entry happens stochastically over time due to regulatory delays. To express the value functions, we need to pin down how state transitions happen regarding generic entry. Thus, we model the evolution of number of generics present as a binomial process following [Ching \(2010\)](#).

Let the number of generics applying to enter market m be n_{em} (which is determined in the first stage). Suppose by period t , $\mathcal{N}_{mt}$ generic entrants have already received approval and launched their products in market m . Thus $n_{em} - \mathcal{N}_{mt}$ entrants have not yet received approval. The probability that of the $n_{em} - \mathcal{N}_{mt}$ remaining entrants, k will receive approval in period t is given by:

$$P_e(k, n_{em}, \mathcal{N}_{mt}, t; \lambda) = \binom{n_{em} - \mathcal{N}_{mt}}{k} \lambda(t)^k (1 - \lambda(t))^{n_{em} - \mathcal{N}_{mt} - k} \quad (7)$$

where $\lambda(t)$ is the probability of a generic entering the market t periods after LOE, and $\lambda = \{\lambda(t)\}_{t=1}^T$. This captures the idea that generics form expectations about entry delays after LOE. We assume the brand drug maker knows n_{em} and total generics that enter for $t = 2$ ($\mathcal{N}_{m2}$).

4.2.2 First-stage

In the first stage, a large number of generics decide whether they want to enter. We assume all generic firms are ex-ante identical, do not receive private error draws for entering and staying out, and do not know their draws of ξ_{jt} conditional on entry.

Generics keep applying for entry until the value from entering no longer exceeds the cost of entry. More precisely, the equilibrium number of generic entrants is n_{em} , determined by the following inequality:

$$V^g(s_{gm0}, n_{em}) \geq \kappa_m^g > V^g(s_{gm0}, n_{em} + 1) \quad (8)$$

where κ_m^g denotes the generic maker's entry cost for molecule-formulation m and s_{gm0} is the initial state of the market from a generic's perspective (i.e., before LOE and generic entry have begun).

This follows the free-entry condition in [Igami \(2018\)](#) and [Ching \(2010\)](#). Intuitively, the value function of entry is declining in the number of entrants – more entrants mean more competition in the market and lower per-period profits. At the equilibrium n_{em} , current entrants can cover their entry cost, but an additional entrant would stop that from happening.

Importantly, the generic's value function incorporates the AG policy function from the second stage when predicting state transitions. Finally, unlike the brand drug maker, the generics do not know $\mathcal{N}_{m2}$.

4.2.3 Model solution

We solve the model by backward induction and recover the equilibrium number of generic entrants and policy function of AGs. Since the generics account for the AG policy function when calculating their value function, and the AG knows n_{em} , all agents are forming ratio-

nal expectations about each other’s choices when making a decision.¹⁸ This means that a policy or environment change that affects the choice probabilities of one agent will have feedback effect on other agents in the market. Our model solution adapts the method of [Igami \(2018\)](#); details of the model solution are given in [Appendix A.1](#).

4.3 Discussion

Interpreting AG’s choice-specific shocks: The choice-specific logit shocks in the AG release decision are used to account for payoff shocks observed by the agent but unobserved to the econometrician. These payoff-shocks could subsume a variety of factors such as manufacturing delays, cost issues, policy/legal uncertainty, inexperience with generic production, alternative lifecycle-extension strategies, organizational/transactional frictions, etc. In our setting, it also subsumes cases where an AG was not released due to a No-AG settlement.¹⁹ This paper does not model the No-AG settlement decision explicitly, partly because it is extremely difficult to identify when such settlements have happened. Our per-period payoff calculations are not affected by No-AG settlements. See [FTC \(2011\)](#) and [Bokhari et al. \(2020\)](#) for a more detailed analysis of such settlements. Finally, the choice-specific payoff shock subsumes the brand maker engaging in other post-LOE strategies (such as product line extension or product discontinuations) that affect its AG release decision. See [Appendix B](#) for a detailed discussion on factors subsumed in these choice-specific shocks.

One concern is that these *i.i.d.* logit shocks are also subsuming shocks that are better thought to be at the molecule-formulation level. For instance, no-AG settlements, brand-firm-level manufacturing constraints, etc. are presumably molecule-formulation level shocks that are being approximated with independent molecule-formulation-quarter-level shocks. We cannot fully account for these unobservables, especially as some of these shocks are not really cost shocks either. We think of our AG entry cost and choice-specific

¹⁸This brings up the question of whether rational expectations is the correct assumption. See [Appendix H.4](#) for quotes from the [FTC \(2011\)](#) showing that generics carefully account for AG entry. See [Appendix 4.3](#) for institutional details suggesting brand drug makers are tracking generic entry.

¹⁹“No-AG settlements” are settlements stemming from Paragraph IV certifications where the first-filer generic firm agrees to delay entry in exchange for assurance that an AG will not be released. Such a No-AG settlement could be related to a Paragraph IV settlement regarding the same molecule-formulation, or a settlement involving a different molecule-formulation but involving the same firms.

shocks as a simple approximation of the true cost. We take care to implement our counterfactuals so as to work around this issue.

Generic launch timing: Recall that we model the entry decision by generics to be a static entry game that happens before loss-of-exclusivity. We choose to model generic entry as a static game instead of a dynamic timing game because regulatory delays create uncertainty for generics about when they can enter the market; as a result, generics usually apply for FDA approval well in advance of the LOE.

Because nearly all generics apply for entry before LOE, a fraction get final approval from the FDA as soon as LOE happens (over 30% in our data). This is accounted for in the generic entry rate calculation; see Section 5.2 for a detailed explanation. The concern that some generics file for entry well after loss-of-exclusivity turns out not to be much of an issue for our setting. Among others, [Starc and Wollmann \(2023\)](#) find that the mean ANDA filing in mature generic markets is 0.21 per drug-year (even lower for non-cartelized markets), and conclude that “entry into (mature) generic markets is typically rare”. [Frank et al. \(2021\)](#) report similar results.

Paragraph IV certifications and settlements: One simplification we make is regarding generics’ decision to file a Paragraph IV certification while applying for entry and any subsequent pay-for-delay settlements. The decisions to file Paragraph IV certification and negotiate any settlements (such as pay-for-delays or no-AGs) are exogenous to our model. More specifically, we think of Paragraph-IV certifications and pay-for-delays determining when a molecule-formulation is open for generic entry. Our model takes this opening date as exogenously given. Regarding first-filer exclusivity, our model assumes each generic assigns some probability that they are the first-filer, or that they have to wait 180 days to enter because a rival achieved the first-filer status instead. These probabilities are reflected in our calculation of generic entry rates.

In general, we do not endogenize decisions surrounding patent lawsuits and settlements, and instead allow drug makers to anticipate them with some probability. Our paper focuses on the periods of and following generic entry. These are simplifications driven by data limitations and computational concerns, and they allow us to tractably model the generic entry decision.

Note that [FTC \(2011\)](#) finds most AGs are released outside the 180-day exclusivity, and often for drugs that did not see any patent challenges; see Appendix H.3. Note also that

FTC (2011) argues AGs are unlikely to deter Paragraph IV certifications.

Information assumptions: We make several information assumptions. First, we assume that the brand drug maker knows how many generics have applied for entry to its molecule-formulation. In reality, FDA informs the brand drug maker of the first-to-file ANDA, but may fail to provide this information on subsequent filers. Full information on each ANDA applications across all generic drug makers becomes public only after they win ANDA approval from the FDA and are listed in the FDA’s orange book. However, it is well known that brand drug makers may know about generic drug makers intention to file ANDAs through a number of other means, including news reports of filings, requests for samples by generic makers to manufacturer the brand product, etc. Thus, while this is a simplifying assumption that improves the tractability of our model, it is also a reasonable approximation of reality. Second, we assume that the AG correctly predicts the number of generic launches in the quarter when LOE happens; this is realistic as the FDA would have announced before LOE which ANDAs have been approved by then.

5 Estimation and Results

We use reduced-form regressions to predict market shares and prices in different states, and estimate generic entry rates from the data. Marginal costs are estimated by matching predicted generic entry numbers with observed generic entry numbers. AG release decision’s entry cost is calibrated and the logit scaling parameter is estimated via Maximum Likelihood. Throughout we set the quarterly discount factor at $\beta = 0.95$ and $T = 32$.²⁰

5.1 Per-period payoffs

To calculate per-period payoffs in different states, we need to predict market shares, prices, and marginal costs in different states. Both market shares and prices are predicted using

²⁰This corresponds to an annual discount factor of 0.81, which is in the range of [0.75,0.95] used by Igami and Sugaya (2022). Some dynamics papers use higher annual discount factor such as 0.95. We choose a lower number because i) we do not account for exit or discontinuation of makers, so the low discount factor may indirectly account for that possibility by downweighting future payoffs more, ii) pharmaceutical industry sees new molecule-formulations experiencing loss-of-exclusivity regularly so there’s a high risk of a product market becoming obsolete in the future. Such risks of product obsolescence is implicitly captured by a lower discount factor.

regressions on observables, specifically those reported in Table 2. This allows us to flexibly predict prices and market shares without making strong assumptions about the demand structure or pricing conduct. In doing so, we assume that all generics charge the same price in a market-quarter, which is a reasonable approximation. The results in Table 2 line up well with findings in the prior literature.

Estimation of the marginal cost parameters is done by matching predicted generic entry numbers to observed generic entry numbers at publicly reported generic entry costs, as explained below.

5.2 Generic entry rates

To operationalize equation (7), we estimate the average generic entry rates $\lambda(t)$ at every quarter-since-LOE. We assume that all generic applicants eventually enter, and thus rule out cases where i) a generic applies for entry and never receives approval and ii) a generic receives approval but chooses to never enter.

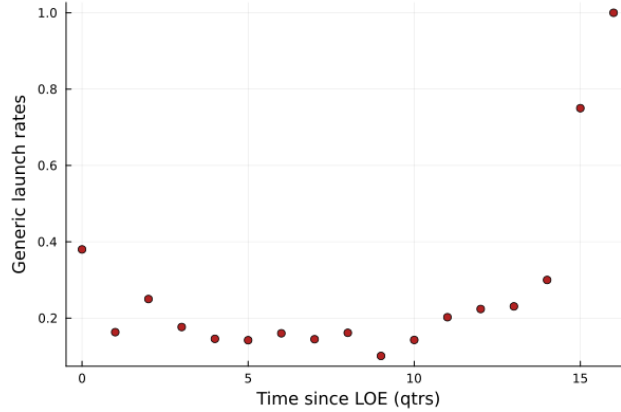
Our estimate of $\lambda(t)$ has to capture the probability of a generic entering t periods after LOE. To do so, we track how many generics are waiting to enter at every quarter-since-LOE, and how many generics enter at every quarter-since-LOE. The ratio of the latter to the former is the generic entry rate at every quarter-since-LOE. The exact description of the calculation is presented in Appendix E.

Nearly all generics apply for FDA approval well in advance of loss-of-exclusivity, and some receive it before or as soon as LOE happens. This is why we see over 30% of the generic entries happening in the quarter of LOE. This also explains why we find generic entry rates to be highest at the beginning of loss-of-exclusivity, and declining over time.

5.3 Marginal costs

We derive approximate bounds on the marginal costs by matching predicted generic entry numbers to observed numbers at publicly reported generic entry costs. Since we do not make assumptions about pricing conduct, we cannot impute marginal costs from observed prices (though as we will see later, we do use the average observed prices in the

Figure 4: Generic entry rate used in model estimation.



Notes: See Table 16 for the calculation of these values.

molecule-formulation to inform the marginal costs of that molecule-formulation).²¹ Instead, we recover marginal costs using observed generic entry numbers. The intuition is that the estimated marginal costs will scale down predicted profits so that at the publicly-reported generic entry costs, the model predicts reasonable generic entry numbers.²²

The estimation is done as follows. We assume that the marginal cost for molecule-formulation m can be approximated by $MC_m = \vartheta \bar{p}_{gm}$, where $\bar{p}_{gm}$ is the average observed price of generics in molecule-formulation m over all periods in our dataset, and ϑ is a scaling parameter. The gain from scaling the average price of generics to estimate marginal costs is that i) generics price close to the marginal cost, so there is information about marginal cost levels from observing their prices, and ii) it allows marginal cost to vary across molecule-formulations.

We isolate molecule-formulations for which AGs do not enter, and set the AGs to have

²¹Dubois and Pakes (2026) use similar data to this paper and suggest that inferring marginal costs of pharmaceuticals using Nash-Bertrand pricing first-order conditions may be problematic. They estimate marginal cost using a dynamic first order condition for advertising, as well as the static Nash-Bertrand pricing first order condition, and the former gave more reasonable results.

²²This suggests that our interpretation of marginal costs is somewhat broader than the cost of producing an additional tablet. Our estimates of marginal costs of operation reflects the cost of operating the factory and production line, maintaining supply chains, transaction costs, etc. It is also the economic cost of launching a product, and so reflects costs beyond accounting costs. For instance, there could be opportunity cost tied to allocating inputs to this product as opposed to elsewhere, or there could be additional constraints facing the managers.

zero probability of entering these markets. For a given generic entry cost κ_g , we try different trial values of ϑ and solve for the equilibrium generic entrants in each market. Using grid-search, we pick the ϑ that minimizes the sum of squared differences between the observed and predicted generic entry numbers. Reported generic entry costs vary between \$3 million and \$15 million, so we set κ_g to each of these bounds and recalculate ϑ . We find that ϑ range from 0.4 to 0.5. We fix ϑ to be the average of these two bounds for our main estimation. Details of the estimation procedure are given in Appendix A.3.

5.4 Estimation of AG entry costs and choice-specific shocks

To model the AG release decision, we need values for the AG entry cost κ_m^{AG} and the logit scaling parameter of the AGs’ choice-specific shocks σ_ε .

We calibrate the AG entry cost κ_m^{AG} . The range of entry costs for generics is widely reported to be between \$3 million and \$15 million. The cost of releasing an AG is likely to be significantly less than the lowest generic entry cost. The generic’s sunk cost of entry mostly comes from having to reverse-engineer the production process of the brand drug and setting up supply chains; this is not a cost the AG release has to incur. Other costs such as production, negotiation, and distribution are captured in the estimated marginal costs.

Thus, we conduct estimation by setting κ_m^{AG} to be between \$250,000 and \$1 million, and set it to be \$250,000 for the counterfactuals. All estimates and counterfactual results are similar across this range of AG entry costs.²³

Next we estimate the logit scaling parameter of the AGs’ choice-specific shocks σ_ε . The parameter is estimated via Maximum Likelihood Estimation; see Appendix A.2 for details. Since equilibrium generic entrants for each molecule-formulation is observed, we only need to solve for the AG choice probabilities for estimation.

We conduct the estimation (and counterfactuals) for non-injectable molecule-formulations in the dataset. This is because very few injectables see AG released, while the non-injectables see most of the AGs released (99 out of 108). This makes it difficult to estimate the AG logit scaling parameter for injectables.²⁴ This leaves us with 182 molecule-

²³We also attempted to estimate κ_m^{AG} and σ_ε jointly, but found that they were poorly identified in practice.

²⁴The lack of AGs in injectables is likely due to the fact AGs are a way for the brand manufacturer to price-discriminate between patients – this works for non-injectables as they are often sold directly to patients, but not for injectables as they are often sold to hospitals and administered by healthcare professionals.

formulations for AG estimation and counterfactuals.

Results of estimation of the logit scaling parameter of AGs are reported in Table 3. We find that the estimated logit scaling parameter is somewhat similar across different values of κ_m^{AG} , and preliminary runs suggest very similar counterfactual results.

Table 3: Results of cost estimation.

κ_m^{AG} (calibrated)	σ_ε	s.e. of σ_ε
250.0	6560.83	398.633
500.0	5121.65	324.499
1000.0	3248.25	170.083

Notes: Calibrated AG entry cost κ_m^{AG} in \$1000s. Per-period payoffs scaled by \$1000s when estimating σ_ε . Estimation done using Maximum Likelihood. Standard errors are reported under “s.e.”.

5.5 Discussion

Price prediction regression: Instead of assuming a pricing conduct such as Nash-Bertrand or Nash-in-Nash bargaining, we use a price prediction regression to predict prices charged by different drug makers in different markets. Any specific pricing conduct assumption is unlikely to be very accurate, since pricing in the pharmaceutical industry is highly complicated.²⁵ Instead, our price prediction regression is a flexible way to model the pricing game without making strong assumptions about the pricing conduct. The advantage is that our predicted per-period profits are not susceptible to misspecifications about the pricing con-

²⁵Other papers which study pharmaceuticals with aggregate data and make a pricing conduct assumption include [Arcidiacono et al. \(2013\)](#), [Dubois et al. \(2022\)](#), and [Starc and Wollmann \(2023\)](#). Unlike these papers there are major differences in our analysis that preclude us from doing the same. [Arcidiacono et al. \(2013\)](#) focus on a small number of molecules, which allows them to leverage additional data and impute rebates. In contrast, our paper considers well over 200 molecule-formulations. [Dubois et al. \(2022\)](#) restrict their attention to sales in the hospital channel for drug sales, while we consider all channels. They assume Nash-Bertrand, which is a more reasonable assumption in their setting. [Starc and Wollmann \(2023\)](#) consider the pharmacy channel, use different datasets compared to this paper, and specifically study price-fixing.

duct assumption.²⁶²⁷ Our price prediction equation is similar in spirit to the price-setting equation in [Maini and Pammolli \(2023\)](#).

Note that in our counterfactuals, we do not change any primitives that would affect the static pricing game. Otherwise, the price prediction equation estimated from the observed data would no longer be a good predictor, as a change in demand primitives should change the underlying pricing patterns. Instead, we limit our counterfactuals to changing parameters that affect dynamic incentives (such as faster generic entry rates and entry restrictions).

The main focus is on entry instead of the static pricing game, which means being able to accurately predict per-period payoffs is sufficient for our analysis.

Another issue is that during this period, some generic makers may have been engaging in price-fixing ([Cuddy \(2020\)](#), [Starc and Wollmann \(2023\)](#)). This alleged conduct occurred from late 2013 onwards; our dataset’s coverage is from 2004-2016. As a result, we rerun the price prediction regression by dropping years 2013 onwards, and obtain very similar results.

Market share prediction regression: We opt to predict market share using an intuitive regression on observables. This regression captures the key substitution patterns and components that will dictate market share. It shows how non-brands steal market share from the brand and from each other, and how the brand appears to have an edge over generics when it comes to attracting consumers. This market share prediction regression helps capture the shape of the per-period payoffs across different states, and serves as an input into our entry model.

The main focus of our paper is on entry instead of the static per-period game, which means being able to reasonably approximate per-period payoffs is sufficient for our analysis. Our market share prediction regression allows us to do that. This also allows us to avoid having to deal with complex random coefficient models and instruments, which in any case will have a hard time capturing the substitution patterns across such a variety of

²⁶Our price prediction regression also gives very reasonable results that line up with the prior literature. For instance, more generics lead to lower prices, but the effect diminishes with the number of generics; the brand drug charges higher prices than generics, and its price stays the same or rises as more generics come in.

²⁷A regression of prices on level of competition may suffer from endogeneity issues from demand shocks. This is less of an issue here than in other settings, because generics cannot simply respond to a demand shock and enter the market - they need to wait years to get FDA approval and launch. Furthermore, we also add in molecule and formulation fixed effects.

molecule-formulations and with such aggregate data.

6 Counterfactuals

In this section we explain how the counterfactuals are conducted, then discuss the results from our counterfactual simulations.

We solve the model by backward induction. For the counterfactuals, we solve for both the number of generic entrants and policy function of AG. A description of the full-solution method is given in [Appendix A.1](#).

For each counterfactual we solve the model, run 200 simulations, then report the average results from the simulations. We run the counterfactuals for 157 of the 182 non-injectable molecule-formulations found in our data. Details of the implementation can be found in [Appendix C](#).

We now briefly discuss profitability in different phases of a molecule-formulation’s post-LOE lifecycle. The early quarters of a molecule-formulation after loss-of-exclusivity are the most profitable for firms. During these periods, there are few firms in the market, resulting in high markups and high economic profits. As time goes by, more generics enter the market, resulting in markups being driven close to zero and low economic profits. This provides another reason why generics want to launch earlier – the earlier phase of the market post-LOE yields the highest profits and so contributes most towards covering the entry cost.

6.1 Entry deterrence in the model

We explain the intuition behind how entry deterrence operates in our model, then show counterfactual results that confirm our intuition. We show that both independent generics and AG can deter each other’s entry, but in different ways. Generics can deter AG release because they make entry decisions earlier; on the other hand, the AG can deter generic entry because it can launch earlier and provide price competition from the early stages of the market.

First, we explain the intuition behind how entry deterrence operates through generics and AGs:

1. An independent generic has an “early-mover” advantage. This is because a generic makes its entry application decision and incurs its fixed cost in the first stage, before the AG. This ability to move earlier means that the generic can deter an AG from entering. Note that the AG has a substantially lower entry cost than independent generics, weakening this entry deterrence effect.
2. An AG has an “early-launch” advantage. This is because the AG can launch its product earlier than independent generics. Since the AG can launch its product early – and during the most profitable phase of the market – it has a strong negative impact on expected lifetime profits of an independent generic, and thus deters generic entry.

To confirm our intuition, we perform the following two counterfactual simulations.

First, we show that generics are able to deter AG release when AGs do not enjoy the launch-timing advantage over generics. We fix the following: i) Generics can launch immediately upon loss-of-exclusivity (at $t = 2$), so that the AG no longer enjoys a launch timing advantage over generics; ii) AG has a low logit scaling parameter so that the AG can launch immediately; iii) AGs have the same entry cost as generics, so that the AG no longer has a cost advantage; iv) AG and generics face the same demand primitives, so that neither enjoys any demand-side advantage over the other. Thus, in this counterfactual, the AG no longer has the early-launch, demand, or cost advantage. The results are summarized in Table 5. We find that across nearly all market-specifications, the AG is not launched.

Next, we show that AG is able to deter generics better when it has a stronger early-launch advantage. To do so, we vary the AG’s early-launch advantage to see how it affects generic entry. First we run simulations where the AG always launches during the LOE period. Next we run simulations where the AG always launches eight quarters after LOE. The results are summarized in Table 6. We find that the AG can deter more generics when it can launch earlier.

6.2 Generic entry rates

We examine the effects of reducing regulatory delays on generic entry. In our counterfactuals, we scale up the probability of generics entering every quarter after LOE (i.e. $\lambda(t)$), and simulate the impact on market outcomes. That is, reducing regulatory delays corresponds

to faster generic entry rates in our model. We find that faster generic entry rates lead to fewer generics applying for entry, lower prices in initial periods, and weakly higher prices in later periods.

Before unpacking the results, we discuss the key economic mechanisms at play. Under faster generic entry rates, there are opposing forces on a generic's entry decision. On the one hand, faster entry rates mean the generic can market their drug earlier, and so has a longer time (until the terminal period) to make profits and cover their entry cost. This force would incentivize more generics to enter. On the other hand, faster entry rates mean that when a generic does enter the market, more of its competitors are already in the market, so it will face greater competition and smaller profits upon entry. This force would disincentivize more generics from entering.

So which force dominates? We can study this by simulating markets under faster generic entry rates.

For the first group of counterfactuals we leave AGs out of the market for clarity. For most market specifications we see that as generic entry rates increase, there are fewer generic entrants; for the remainder, there is no change in total generics. This means that the entry disincentive from earlier competition generally dominates the entry incentive from longer operation time in the industry.²⁸ The results are summarized in Table 7. An example for a single molecule-formulation is given in Table 4.

Our counterfactual simulations help shed light on why the disincentive effect dominates. Take a molecule-formulation where, if generics can enter immediately, then there will be three generic entrants. If an additional generic were to enter, the prices would be driven down so low that the additional generic would not be able to cover its entry cost. Now, let's take the same market, but with a slower entry rate. The first three generics will still choose to enter, but their entries are now more spaced out over time. What about an

²⁸To see why, we can look at the per-period prices (and profits) in the industry. The earliest periods are the most profitable, because few nonbrand products are in the market, resulting in high markups. As more generics enter, these markups are driven down to zero, so that in the long run, the price is nearly equal to marginal cost. This means that after a certain number of periods, generic firms no longer make economic profit that can go towards covering their entry cost. With faster generic entry rates, price drops to marginal cost much faster, meaning there are far fewer quarters when generics can make economic profit, because there will be more competitors in the market right from the beginning. Thus, while it is true that generics can operate in the market longer due to being approved earlier, most of that time is going to be spent making near-zero economic profit.

additional fourth entrant? If the fourth entrant enters, then in the long-term the prices will be very low as in the immediate case. But in the short term, the entrant might get a lucky early approval, enter the market in its early periods when there is little competition, and make large profits. Thus, for the fourth entrant, the prospects of entering the market have improved with the slower entry rate. Slowing down the entry rate creates an element of luck where an additional entrant can make large profits in the early periods of the market.

Next, we rerun the counterfactuals but allow the AG to be released. Similar to before, for many market specifications, as generic entry rates increase, there are fewer generic entrants. However, with AG, the decline in generics as entry rate increases is smaller; that is, as entry rate increases, the number of generics declines by a smaller amount with AG presence than without AGs. In fact, there are more market specifications for which there is no change in number of generics. The results are summarized in Table 8. The intuition is that, with the AG present during the early periods of the market, the gain from receiving a lucky early approval is much smaller. Thus a marginal entrant has less incentive to take a gamble and enter when generic entry rates are slower.

While faster entry rates lead to fewer generics, this has mixed effects on prices – prices are lower in the early periods of the market but are often higher in the later periods. Prices are lower in the early periods because, even though there are fewer generic entrants, most of them get released very quickly into the market after LOE. However, the fact that there are fewer generic entrants in total means that prices are often higher in the later periods of the market.

Table 4: Market outcomes with different generic entry rates for one molecule-formulation. The AG is prevented from entering the market for expositional clarity.

Cases	Total generics	Mean generic price (early periods)	Mean generic price (later periods)
Estimated	7.0	0.75	0.35
Estimated x 2	6.0	0.5	0.42
Immediate entry	6.0	0.41	0.42

Notes: Simulation conducted for a representative molecule-formulation. “Estimated x n ” refers to the estimated generic entry rates multiplied by n . “Immediate entry” refers to all generics entering the market immediately after loss of exclusivity.

6.3 Ban on Authorized Generic

We examine if AGs (and such fighting-brand strategies) should be encouraged or discouraged by policymakers. To do so, we simulate market specifications under allowing AGs to be released versus a ban on AGs. We limit our counterfactuals to molecule-formulations that saw an AG released in the data, to assuage worries about unobserved molecule-formulation shocks that may have deterred AG release. We find that a ban on AGs leads to weakly more generic entrants – and sometimes multiple additional new generics – but has mixed effects on overall prices.

In our counterfactuals we find that, when AGs are banned, nearly all market specifications see an increase in the number of generics. In particular, around 25% of the markets see multiple additional new generics entering. For the large majority of market-specifications, the AG ban leads to higher prices in the early periods of the market. If the AG was deterring multiple generics from entering, prices are lower in the later periods of the market with an AG ban. The exact effect on number of generics and prices depends on the specific demand and supply parameters. The results are summarized in Table 9.

We first explain the impact of an AG ban on generic entrants. There are two opposing forces operating on different phases of the market’s timeline after loss-of-exclusivity:

1. Early AG launch: Without an AG ban, the AG is released early in the market, so there is always an additional competitor in the market from the early quarters. Without the AG, the generics face lower competition in the initial periods as all the generics stochastically enter over time. Therefore, under an AG ban, prices are generally higher in the early periods of the market.
2. Greater generic entry: An AG ban leads to more generic entrants. If there are two or more additional generic entrants, then prices are lower in the later periods of the market.

The effect on later-period prices depends (among other things) on how many generics are deterred by the presence of an AG. If an AG ban leads to multiple new generics entering, then the resulting increase in competition more than offsets the missing price competition from the AG. As a result, in these markets the AG ban leads to lower prices in later periods. However, if an AG ban leads to one or no new additional generic entrants, then an AG ban leads to similar prices in the later phase of the market.

Why can an AG ban lead to more than one new generic entering? This is because the AG has a larger negative impact on rivals' lifetime profits than an independent generic due to its early-launch advantage. Since the AG is released early, it has a negative impact on every generic and particularly during the most profitable early periods of the market. If the AG is banned, not only is there one fewer competitor, but it is also one less competitor during the most profitable periods in the market. This results in a significant increase in expected generic profits compared to one fewer independent generic. In turn, this can incentivize multiple new generics to enter than if the AG had been present.

Exactly how many independent generics are deterred by the presence of the AG in our simulations also depends on market-specific factors, particularly generic entry costs.

7 Conclusion

In the US pharmaceutical industry, brand drug makers can compete with generics by releasing an Authorized Generic, which is identical to the brand drug but without the brand label attached. Using quarterly drug sales and revenue data on the US for 2004-2016, we estimate a structural model of pharmaceutical demand, generic entry, AG release, and pricing. We build a two-stage supply model of static generic entry followed by a dynamic AG release timing game. The model is solved using backward induction to recover equilibrium generic entrants and policy function of AG release. We estimate this model to recover supply-side parameters.

We simulate market equilibrium under different environments. First, we show how deterrence works in our model - generics deter through being able to file for entry earlier (early-mover advantage) while AGs can deter because they can launch the product earlier. Next, we find that a faster entry rate leads to weakly fewer generic entrants. This is because greater competition upon launch offsets the advantage from an early launch. Finally, we show that a ban on AGs leads to weakly more generic entrants and mixed effect on overall prices. The latter happens because while there is increased price competition from weakly more generic entrants in the later stages of the market, it is counteracted by lack of an additional competitor through the AG in the early stages of the market. The exact effect depends on how many generics is being deterred by the AG; for a number of market specifications, we find that the AG may deter multiple generics. In general, policies encour-

aging or banning AGs are offset by generic response, attenuating the impact on consumer welfare.

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Online Appendix

A Model solution and cost estimation

A.1 Solving the full model

We solve the model by backward induction and recover the equilibrium number of generic entrants and policy function of authorized generics. We adapt the approach from [Igami \(2018\)](#) to our setting.

Note that the AG maker knows the total generics approved in $t = 2$, i.e. $\mathcal{N}_2$, but can only form probabilistic beliefs about future generic launches. In contrast, generic makers only know the equilibrium generic entrants n_e , not $\mathcal{N}_2$.

However, the CCP of the AG varies based on the exact value of $\mathcal{N}_2$, since they suggest different probability distributions over future generic launches. That is, the CCP of AG varies by $\mathcal{N}_2$.

The CCP of the AG is used to calculate the value function of the generics in stage 1. While the generics don't know the exact value of $\mathcal{N}_2$, they can form probabilistic beliefs about it using the FDA approval rates.

Thus, for the model solution, first we solve for the CCP of the AG and construct the corresponding generic value function for each $\mathcal{N}_2$. Then each generic value function (at the initial state s_1) is weighted by the probability of the corresponding $\mathcal{N}_2$ being realized. The resulting weighted generic value function is then used for the first stage.

The solution method is as follows:

1. For a given number of equilibrium generic entrants $\bar{n}$:
 - (a) For each $\mathcal{N}_2$ between 0 to $\bar{n}$:
 - i. Calculate the authorized generic's state transition matrices as determined by $\bar{n}$, T , FDA approval rates, and $\mathcal{N}_2$.

- ii. Solve for policy function of authorized generic by backward induction from terminal period T . The CCP gives the probability of AG being released in every state of the world from $t = 1$ to T .
 - iii. Use the CCP, (7), T , and FDA approval rates to calculate the state transition matrices for generics.
 - iv. These state transition matrices are combined with per-period profits to construct value functions of generics.
 - v. Separately, calculate the probability of this $\mathcal{N}_2$ being realized using data on FDA approval rates.
- (b) For the initial state s_1 , find the generic value function by taking the probability-weighted average of the generic value functions for each $\mathcal{N}_2$ at the state s_1 . The probability of each $\mathcal{N}_2$ is being used as weights in this weighted average.
- (c) Denote this value function as $V(\bar{n})$ (other arguments suppressed).
2. Repeat the above steps for $\bar{n} = 1, \dots, N$ where N is a large integer.
 3. The equilibrium number of generics n_e is such that $V(n_e) \geq \kappa_m^g$ and $V(n_e + 1) < \kappa_m^g$. The value function is decreasing in $\bar{n}$.
 4. For this n_e re-solve the model to get the equilibrium policy function of authorized generics conditional on n_e and every possible $\mathcal{N}_2$.

Note that during simulations for a given market, we run many simulations for each possible $\mathcal{N}_2$, then report probability-weighted averages of each summary statistic (such as AG release fraction, AG release time, mean prices, etc.). The probability of each $\mathcal{N}_2$ is being used as weights in this weighted average. We rescale probabilities if some $\mathcal{N}_2$ see no AG released.

A.2 AG release estimation

For AG release estimation, we can skip a large chunk of the full solution method. From the data we observe n_{em} for each market m as well as $\mathcal{N}_{m2}$.

For a given value of parameters:

1. We solve for policy function of authorized generic by backward induction from terminal period T . The CCP gives the probability of AG being released in every state of the world from $t = 1$ to T .
2. The choice probabilities are used to calculate the likelihood.

This process is repeated for different trial values of parameters until the objective function is minimized.

We estimate the model by Maximum Likelihood.

The contribution to the likelihood function from the observation of molecule-formulation m in period t is given by:

$$f(AG_{mt}|s_{mt}, \sigma_{\epsilon}) = pr(AG_{mt} = 1)^{1(AG_{mt}=1)} pr(AG_{mt} = 0)^{1(AG_{mt}=0)}$$

In our estimation we set $T = 32$ quarters. However, many molecule-formulations are followed for much less than 32 quarters in our data. Thus, let $T_{max}(m)$ be the maximum number of quarters a molecule-formulation is followed.

The likelihood for observations of molecule-formulation m is given by

$$\mathcal{L}_m(\sigma_{\epsilon}) = \prod_{t=0}^{T_{max}(m)} f(AG_{mt}|s_{mt}, \sigma_{\epsilon})$$

The joint likelihood over all molecule-formulations is given by

$$\mathcal{L}(\sigma_{\epsilon}) = \prod_m \mathcal{L}_m(\sigma_{\epsilon})$$

During estimation, we maximize the log-likelihood to recover the parameter σ_{ϵ} .

$$\arg \max_{\sigma_{\epsilon}} \log \mathcal{L}(\sigma_{\epsilon})$$

All other parameters informing the policy function in the likelihood function are either calculated offline or calibrated.

A.3 Estimation of marginal costs

Fix some trial value of generic entry cost κ_g across all molecule-formulations. We try $\kappa_g = \$3$ million and $\$15$ million, which are the publicly reported lower and upper bounds on generic’s sunk entry costs. (Intuitively, the $\$3$ and $\$15$ million provide upper and lower bounds on the marginal cost scaling parameter ϑ respectively, as we explain below.) For a given value of κ_g , we pick a grid of values for ϑ at 0.1 intervals between 0 and 1. For each value on the grid, solve for the equilibrium number of generic entrants.

The objective function is the sum of squared differences between the observed generic entrants in each market and the predicted entrants. One issue is that the model cannot predict less than zero entrants for a market (i.e. the outcome variable is censored at 0) and so the objective function may be minimized by predicting zero entrants for some markets. To tackle this problem, the objective function also includes a “censoring penalty”, where if a market is predicted to have zero generic entrants, it is assigned the censoring penalty value instead of the difference between observed and predicted entrants. This is to prevent the estimation from assigning zero entrants to lots of markets to minimize the objective function. In our estimation, we try censoring penalties of 3 and 4 and obtain similar results. We find the objective function to be convex over our grid of ϑ values.

Returning to the estimation, for a given κ_g , we calculate the objective function for every value on the grid. We pick the value that minimizes the objective function. This gives us the ϑ for that κ_g . We repeat this process for the different κ_g mentioned previously.

Note that for the marginal cost estimation, we pick only molecule-formulations, and set the probability of AG release to be zero. This is because the probability of AG release cannot be calculated without knowing the marginal cost.

A.4 Deterrence of first generic entrant

For our paper, we choose only molecule-formulations which saw generic entry in our dataset. What are the implications of this?

In cost estimation, we may be ignoring markets where the first generic never entered because of the threat of AG entry, or no generic filed a Paragraph-IV certification because of AG. Note that [FTC \(2011\)](#) argues it is very unlikely AGs deter Paragraph-IV certifications.

This then means in the counterfactual, we are implicitly ignoring potential molecule-

formulations which would have a Paragraph-IV certification or a first generic entering if AG is banned. In this paper we focus on markets which saw any generic entry in the data, and so cannot predict which markets may have seen generic entry in the case of an AG ban.

In our counterfactual simulations for the AG ban, we also drop molecule-formulations where AG release results in the equilibrium number of generics to zero. It was not clear to us how else to deal with this, because we are unsure if our counterfactual predicts zero generic entry because the calibrated generic entry cost is too high for that molecule-formulation.

B Discussion on AG's choice-specific shocks

During estimation, these logit shocks are used to reconcile why we observe firms in similar states of the world make different choices. This is a common methodology used in dynamic discrete choice models with dollar-valued payoffs. Papers which have unitless payoff functions instead normalize the logit scaling parameter to 1.0 and estimate all other structural parameters relative to it, as in [Rust \(1987\)](#).

A more detailed list of such factors is as follows. First, this could be driven by manufacturing delays, organizational frictions, and contracting frictions (if AG production is being outsourced). It may be that the loss-of-exclusivity was unanticipated or there was uncertainty surrounding it, which meant that the brand maker was unprepared for the loss-of-exclusivity when it happens. There could be policy uncertainty such as how the FDA would approve or regulate the new generic entrants, or how a lawsuit might be resolved, that may add further uncertainty to the release decision. It could be that the brand maker is engaging in other generic-deterring strategies (such as product-line extensions) which affects its AG timing decision. The brand drug maker may not have any generic subsidiaries, which makes it very difficult to release an AG, because marketing a nonbrand drug versus a brand drug entail very different skillsets. The marketing of a brand drug focuses on advertising and detailing, negotiating formulary positions by insurer, etc. The marketing of a generic drug focuses on extreme cost efficiency, winning pharmacy auctions ([Cuddy \(2020\)](#)), etc. These are very different skillsets, and for a brand drug, it may not be worth the hassle of getting into the generics business just to release an authorized generic. It could instead outsource its production to another generic firm (for instance, Prasco and Greenstone specialize in bringing authorized generics for other firms into the market) but that involves negotiation cost and profit-sharing.

C Counterfactual results

For implementing the counterfactuals, we divide the 240 molecule-formulations into “small”, “medium”, and “large” markets. The “small” markets are assigned generic entry cost of \$2 million, “medium” markets are assigned generic entry cost of \$5 million, and “large” markets are assigned generic entry cost of \$15 million .

Now we explain how we classify markets into these 3 groups. The general problem is that for the lowest entry cost of \$5 million, a number of molecule formulations predict 0 generic entrants or more than 22 generic entrants (the highest number observed in our dataset).

As a result, for each molecule-formulation, we try the three generic entry costs mentioned previously. A molecule-formulation is assigned to the group with the largest costs such that the model predicts positive generic entrants less than 15 and more than zero.

For 25 of the 182 non-injectable molecule-formulations our model does not predict positive generic entrants, and we drop those markets from our counterfactuals. See [Appendix A.4](#) for a brief discussion on this. In the end, we have 15 molecule-formulations in the “small” markets, 100 molecule-formulations in the “medium” markets, and 42 molecule-formulations in the “large” markets.

Note that we think of counterfactuals under different market specifications as mainly checking how our model predictions vary for different realistic parameter values.

A note on terminology: throughout this section “Early periods” refers to first 4 quarters, and “Later periods” refers to last 4 quarters (the terminal period is quarter 32).

C.1 Early-mover vs early-launch

Table 5: Distribution of outcomes across market-specifications.

Percentiles	AG release fraction
10%	0.0
20%	0.0
30%	0.0
40%	0.0
50%	0.0
60%	0.0
70%	0.0
80%	0.0
90%	0.0

Notes: For this counterfactual we fix the following: i) Generics can launch immediately upon loss-of-exclusivity (at $t = 2$), so that the AG no longer enjoys a launch timing advantage over generics; ii) AG has a low logit scaling parameter so that the AG can launch immediately; iii) AGs have the same entry cost as generics, so that the AG no longer has a cost advantage; iv) AG and generics face the same demand primitives, so that neither enjoy any demand-side advantage over the other. We find that AG almost never launches.

Table 6: Distribution of outcomes across market-specifications.

Percentiles	Difference in generics
10%	-1.0
20%	-1.0
30%	-1.0
40%	-1.0
50%	-1.0
60%	-1.0
70%	-1.0
80%	-1.0
90%	-1.0

Notes: We vary the AG's early-launch advantage to see how it affects generic entry. First we run simulations where the AG always enters in the period of LOE. Next we run simulations where the AG always enters 8 quarters after LOE. "Difference in generics" is the equilibrium generic entrants when the AG launches early minus the equilibrium generic entrants when the AG launches late. We find that across most market-specifications, the AG can deter more generics when it can launch earlier.

C.2 Faster generic approval rates

Table 7: Distribution of outcomes across market-specifications with estimated versus immediate generic approval. The AG is prevented from entering for expositional clarity.

Percentiles	Difference in generics	Difference in generics price (early periods)	Difference in generics price (later periods)
10%	0.0	0.103	-1.832
20%	0.0	0.215	-0.539
30%	1.0	0.359	-0.333
40%	1.0	0.483	-0.222
50%	1.0	0.62	-0.145
60%	1.0	0.863	-0.075
70%	1.0	1.241	-0.02
80%	2.0	2.402	0.0
90%	2.0	5.056	0.0

Notes: “Difference in no. of generics” is the equilibrium generic entrants under estimated generic approval rates minus the equilibrium generic entrants under immediate generic approval. Other variables similarly defined. We find that faster generic approval rates lead to fewer generics entering the market.

Table 8: Distribution of outcomes across market-specifications with estimated versus immediate generic approval.

Percentiles	Difference in generics	Difference in generics price (early periods)	Difference in generics price (later periods)
10%	0.0	0.061	-1.197
20%	0.0	0.113	-0.357
30%	0.0	0.217	-0.209
40%	1.0	0.341	-0.12
50%	1.0	0.462	-0.075
60%	1.0	0.641	-0.011
70%	1.0	0.883	0.0
80%	1.0	1.716	0.0
90%	2.0	3.848	0.0

Notes: “Difference in no. of generics” is the equilibrium generic entrants under estimated generic approval rates minus the equilibrium generic entrants under immediate generic approval. Other variables similarly defined. We find that faster generic approval rates lead to fewer generics entering the market.

C.3 Ban on AG

Table 9: Distribution of outcomes across market-specifications.

Percentiles	Difference in no. of generics	Difference in generics price (early periods)	Difference in generics price (later periods)
10%	-2.0	-0.608	-0.0
20%	-2.0	-0.332	0.0
30%	-1.0	-0.23	0.0
40%	-1.0	-0.176	0.0
50%	-1.0	-0.101	0.0
60%	-1.0	-0.058	0.0
70%	-1.0	-0.012	0.0
80%	-1.0	-0.001	0.036
90%	-1.0	0.04	0.217

Notes: “Difference in no. of generics” is the equilibrium generic entrants when AG is allowed minus equilibrium generic entrants when AG is banned. Other variables similarly defined. We find that banning AG generally leads to more generic entrants.

Table 10: Distribution of outcomes across market-specifications.

Difference in no. of generics	Proportion of Molecule-Formulations
-2	0.25
-1	0.75

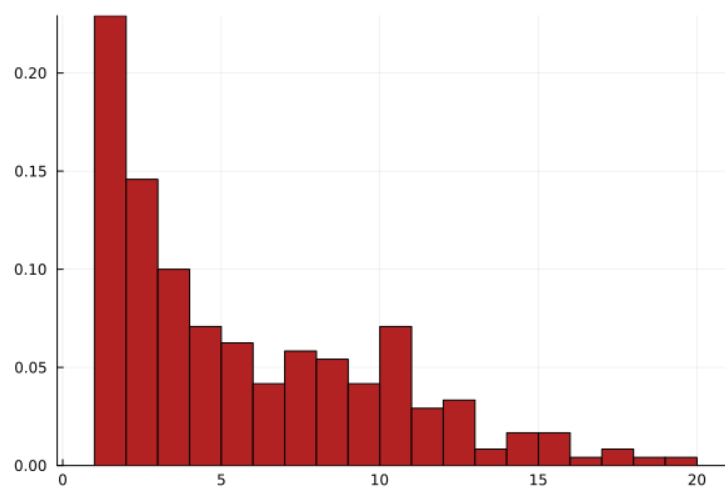
Notes: “Difference in no. of generics” is the equilibrium generic entrants when AG is allowed minus equilibrium generic entrants when AG is banned. Other variables similarly defined. We find that banning AG generally leads to more generic entrants.

D Additional summary statistics

Delay	Count
0.0	58
1.0	10
2.0	6
3.0	2
4.0	2
5.0	3
6.0	2
8.0	1
9.0	1
10.0	1
11.0	1
12.0	2
14.0	1
15.0	2
17.0	1
19.0	2
20.0	2
21.0	1
22.0	2
27.0	1
38.0	1
41.0	1

Table 11: AG Delay table. This shows how many quarters after loss-of-exclusivity the AGs were released. “Delay” is the number of quarters after loss-of-exclusivity. “Count” is the number of AGs released for the corresponding quarters since loss-of-exclusivity.

Figure 5: Histogram of total generic entry by molecule-formulation



Notes: This distribution has a 25th percentile of 2, median of 4, 75th percentile of 8, and 90th percentile of 11. There are 240 molecule-formulations.

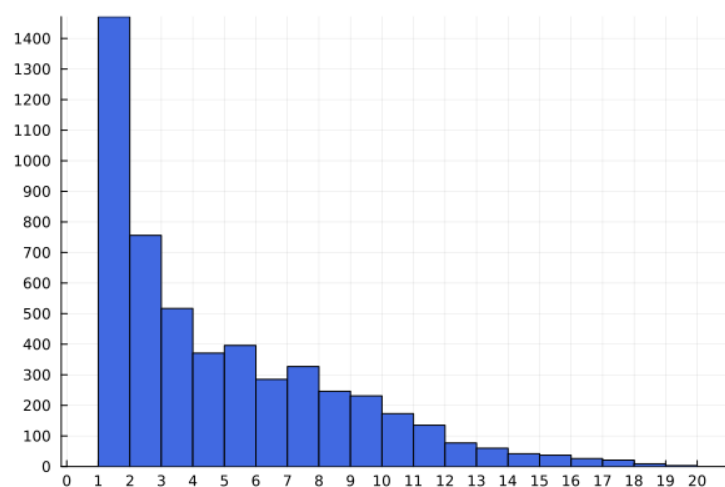


Figure 6: Histogram of count of generics present by molecule-formulation-quarter.

Table 12

	Percentiles	10%	25%	50%	75%	90%
Generic revenue		0.01	0.13	0.79	3.13	9.2
AG revenue		0.1	0.63	3.46	8.67	24.98
Brand revenue		0.18	0.87	4.41	22.08	99.1
Mean brand revenue		0.35	1.2	4.94	19.84	67.81
Mean generic revenue		0.29	0.82	2.35	5.36	12.33
Mean AG revenue		0.32	1.27	4.32	12.15	25.19
Mean brand within molform mktshare		3.51	6.21	15.61	34.92	69.76
Mean generic within molform mktshare		7.96	11.05	18.54	31.91	51.59
Mean AG within molform mktshare		4.19	11.57	22.04	36.01	49.7
Generic total lifetime revenue by product		0.26	1.73	11.06	54.78	184.7
AG total lifetime revenue by product		3.97	10.99	54.45	201.78	444.5

Notes: Values are in millions of dollars.

Table 13: Comparing brand’s quarterly revenue to AG’s total lifetime revenue.

	10%	25%	50%	75%	90%
Equivalent Exclusivity	0.150853	0.510536	1.44618	3.16599	5.00623

Notes: We calculate for each AG their total observed revenues over the data sample, which we henceforth refer to as "observed lifetime revenue". We restrict to AGs which are present for 10 or more quarters. Then, for each AG, we divide its observed lifetime revenue with the corresponding brand’s representative quarterly revenue. We calculate the brand’s representative quarterly revenue as the average of its quarterly revenues in the 3 quarters before LOE. This then gives “equivalent exclusivity in quarters” – how many quarters of brand’s quarterly revenue is raised by an AG over its lifetime. (This is a lower bound as some AGs keep making revenue after the end of my sample).

In Table 14 we compare prices, market shares, and revenues between AGs and generics in the same molecule-formulation-quarter. For each molecule-formulation-quarter where at least one AG and one generic are present, we calculate the difference between the AG’s price, market share, and revenue and the average price, market share, and revenue of the generics in that molecule-formulation-quarter. We then report the distribution of these differences across all molecule-formulation-quarters.

Table 14: Comparison of prices, market shares, and revenues between AGs and generics in the same molecule-formulation-quarter.

	10%	25%	40%	50%	75%	90%
AG minus Generic Price	-1.535	-0.228	-0.025	0.027	0.528	2.286
AG minus Generic Share	-24.857	-6.692	-0.282	4.598	21.824	40.342
AG minus Generic Revenue (millions)	-4.083	-1.42	-0.334	0.194	4.509	11.928

Notes: “AG minus Generic Price” is the AG’s price minus the average price of generics in the same molecule-formulation-quarter. “AG minus Generic Share” is the AG’s market share minus the average market share of generics in the same molecule-formulation-quarter. “AG minus Generic Revenue” is the AG’s revenue minus the average revenue of generics in the same molecule-formulation-quarter.

Table 15: Regressions on market shares and prices for injectable molecule-formulations.

	log(market share)	log(price)
	(1)	(2)
Count of non-brands	-0.216*** (0.024)	-0.310*** (0.018)
Count of non-brands squared	0.009*** (0.002)	0.011*** (0.001)
Brand	0.463*** (0.067)	0.044 (0.044)
Brand x qtrs since LOE	-0.011*** (0.002)	0.001 (0.002)
Authorized generic	-0.023 (0.079)	0.152* (0.063)
Brand $\times$ Count of non-brands	-0.071 (0.041)	
Brand $\times$ Count of non-brands squared	-0.001 (0.005)	
Brand $\times$ Count of generics		0.174*** (0.012)
Brand $\times$ AG presence		1.138*** (0.075)
Molecule-Formulation Fixed Effects	Yes	Yes
Obs.	4,278	4,278

Notes: The variable “qtrs since LOE” is the number of quarters since loss-of-exclusivity. “AG presence” is an indicator for whether an authorized generic is present in the market. “Generic” and “Brand” are indicators for generic and brand drugs respectively.

E Calculating generic entry rate for model solution

We estimate generic entry rates $\lambda(t)$ at every quarter-since-LOE. We assume that all generic applicants eventually enter, and thus rule out cases where i) a generic applies for entry and never receives approval and ii) a generic receives approval but chooses to never enter.

The overall intuition is as follows. Using our data, we want to track how many generics are waiting to enter at every quarter-since-LOE, and how many generics enter at every quarter-since-LOE. The ratio of the two gives us the generic entry rate at every quarter-since-LOE.

The estimation of generic entry rates is done as follows. For each molecule-formulation and quarter-since-LOE, we calculate the number of generics waiting to enter, i.e. final number of generics observed in that molecule-formulation minus the current number of generics present in the molecule-formulation. We sum this number across all molecule-formulations at a given quarter-since-LOE to get the remaining entrants for every quarter since loss-of-exclusivity in our dataset. Then we find the total generic entry that happens in every each molecule-formulation and quarter since loss-of-exclusivity.²⁹ We sum this number across all molecule-formulations at a given quarter-since-LOE to get the total generic entry for every quarter-since-LOE in our dataset. Taking the ratio of total generic entry by remaining entrants at every quarter-since-LOE gives us the average generic launch rates at every quarter-since-LOE. The exact calculation is shown in Table 16.

In our calculation of generic entry rates, we limit the generic entries to those happening on or before 17 quarters.³⁰ Given how we calculate generic entry rates, this mechanically imposes that as a market moves very close to 17 quarters, the generic entry rate rises up to 1.0. For instance, in the 17th quarter, the total generic entry must equal the total remaining

²⁹A generic maker’s entry into a molecule-formulation is identified as the earliest quarter that maker is recorded with non-zero sales in our dataset.

³⁰Our main worry is that entry happening after 17 quarters could involve generics who applied well after loss-of-exclusivity; filings happening after LOE could be for other reasons not modeled in our framework. [Starc and Wollmann \(2023\)](#) find that the majority of ANDA filings are approved within 4 years of filing (see Appendix F.4 in this paper), hence we limit our analysis to 17 quarters, and robustness checks with 24 quarters are very similar. It turns out to matter little for our results because there are few generic entries happening after 17/24 quarters in any case.

generics.

Nearly all generics apply for FDA approval well in advance of loss-of-exclusivity, and some receive it before or as soon as LOE happens. This is why we see over 30% of the generic entries happening in the quarter of LOE. This also explains why we find generic entry rates to be highest at the beginning of loss-of-exclusivity, and declining over time, before rising back up close to the 17th quarter.

From our data, over our sample period, there are 771 generic entries in total. We can think of this as 771 generics waiting to enter at the time of LOE, and their actual entries are staggered out over time. As more entries happen, the pool of generics waiting to enter shrinks.

- At LOE, there are 771 generics waiting to enter, and 293 generics enter at that period, so the entry rate is $293/771 = 0.38$.
- In the next period, the number of generics waiting to enter is $771 - 293 = 478$. The number of generics which enter from that pool of 478 is 78, so the entry rate is $78/478 = 0.163$.
- In the next period, the number of generics waiting to enter is $478 - 78 = 400$. The number of generics which enter from that pool of 400 is 100, so the entry rate is $100/400 = 0.25$.
- In the next period, the number of generics waiting to enter is $400 - 100 = 300$. The number of generics which enter from that pool of 300 is 53, so the entry rate is $53/300 = 0.177$.
- And so on.

In our calculation of generic entry rates, we limit the generic entries to those happening on or before 17 quarters.

Given how we calculate generic entry rates, this mechanically imposes that as a market moves very close to 17 quarters, the generic entry rate rises up to 1.0. This is because as we approach 17 quarters, there are fewer generics remaining, and so few entries can lead to a high entry rate. In the last quarter (17 quarters after LOE), the entry rate is 1.0 because all remaining generics must enter. For expectation formation by generics, this effectively

implies that all generics that have not entered by 17 quarters after LOE will enter at that time.

Table 16: Calculation of generic entry rates $\lambda(t)$ for model estimation.

Time since LOE	Generic entries	Remaining generics	Generic Entry Rate
0.0	293.0	771.0	0.38
1.0	78.0	478.0	0.163
2.0	100.0	400.0	0.25
3.0	53.0	300.0	0.177
4.0	36.0	247.0	0.146
5.0	30.0	211.0	0.142
6.0	29.0	181.0	0.16
7.0	22.0	152.0	0.145
8.0	21.0	130.0	0.162
9.0	11.0	109.0	0.101
10.0	14.0	98.0	0.143
11.0	17.0	84.0	0.202
12.0	15.0	67.0	0.224
13.0	12.0	52.0	0.231
14.0	12.0	40.0	0.3
15.0	21.0	28.0	0.75
16.0	7.0	7.0	1.0

F Miscellaneous

F.1 Links

1. Bill introduced to ban AGs: <https://www.mmm-online.com/home/channel/regulatory/bills-would-ban-authorized-generics/>
2. The FDA list on AGs can be found here: <https://www.fda.gov/media/77725/download>.
3. The online source by AG producers can be found here: <https://www.authorizedgenerics.com/product-finder/>
4. The FDA list on Paragraph IV lawsuits can be found here: <https://www.fda.gov/media/133240/download>.

F.2 Names of brand producers launching AGs

Names of brand producers launching AGs in our dataset: Pfizer, Novartis, Sanofi Aventis, Valeant, Abbvie, Boehringer Ingel, Janssen Pharma, Astrazeneca, Concordia pharma, GSK pharma, Leo Pharmaceuticals, Merck, Takeda, Teva, Lilly, Meda, Oak, Shionogi, Actavis, Bayer, Bristol-Myers Squibb, Celgene, Daiichi Sankyo, Eisai, Endo, Genzyme, Impax, Lundbeck, Merz, Mission, Ortho, Par, Parapro, Salix, Savient, Schering, Sebela, Symplmed, The medicines co, VIIV healthcare.

F.3 Comparison with [Fowler et al. \(2023\)](#)

One difference between [Fowler et al. \(2023\)](#) and our findings is they find AG incidence is just 6% of the markets, while we find nearly 44%. We believe this is mainly because they define product and possible markets for AG release (“at risk markets”) more broadly than us. In particular, differences in AG entry rate arise due to:

1. Definition of product: We consider a product market to be “molecule-formulation”, where formulation is either oral, injectable, or everything else. [Fowler et al. \(2023\)](#) define a product to vary by strength and more detailed dosage forms. As a result, their number of markets is much larger than ours.
2. At-risk markets: We consider a market to potentially see AG entry (at-risk market) if we find that it has seen at least one generic entry (and that first generic entry happened after 2004). Instead, [Fowler et al. \(2023\)](#) considers at-risk products to be “number of products approved during or after 1985 that had not been withdrawn or discontinued and that still had patent or exclusivity protection in each year from 2010 to 2019”. This is a very broad definition. Since AG release is only considered when a generic enters, our definition is consistent with our research question.
3. Undercounting AGs: [Fowler et al. \(2023\)](#) mentions that for data issues “... led to conservative estimates of the incidence of authorized generic entry”
4. Difference in yearly coverage: We look at 2004-2016, while they consider 2010-2019.

Other findings from [Fowler et al. \(2023\)](#) match our paper closely.

F.4 Comparison of generic approval rates with [Starc and Wollmann \(2023\)](#)

Another paper that measures FDA approval rates of generics is [Starc and Wollmann \(2023\)](#). Their setting and data is different from ours. [Starc and Wollmann \(2023\)](#) studies generic application and entry for markets where the loss-of-exclusivity has happened well in the past. They also have data on when the generic filed their ANDA, and so they can exactly measure the time between filing and approval. They report the distribution of "regulatory delays" in years and report it in a graph; numerically they find the following distribution:

- 1 year: 0.1
- 2 year: 0.34
- 3 year: 0.34

- 4 year: 0.15
- 5 year: 0.07
- 6 year: 0.04

(These figures are read off a graph and so aren't exact.)

Compared to our estimates, they find much slower ANDA approval rates. We are not sure why that is. One reason could be that we are modeling markets which face loss-of-exclusivity. For some of these markets, the loss-of-exclusivity is anticipated, and so generics may start filing in advance. As a result, they arrive on the market much sooner compared to the setting in [Starc and Wollmann \(2023\)](#), where generics notice a cartel formation/high prices due to cartel formation and then apply for ANDA approval.

As a result we think that while [Starc and Wollmann \(2023\)](#) calculates approval rates that are correct for their setting, they might be too slow for our setting.

F.5 Terminology about patent challenges

There are sometimes patent-related lawsuits and settlements involved around generic entry and AG release, which we now discuss briefly.

A “Paragraph IV certification” lawsuit by the generic maker claims that the brand drug’s patent is invalid or will not be infringed upon by the generic maker’s product. A generic maker can file a Paragraph IV to vacate patent protection for a brand drug and enter the market.

The Hatch-Waxman Act set up a “first-filer exclusivity” provision where the first generic to file for Paragraph IV certification enjoys a 180-day period after entry when other generics cannot enter; thus, a rival becoming the first-filer stops other generics from entering the market for 180 days.

When a generic files a Paragraph IV certification, the brand drug maker can agree to pay them directly to abandon the patent challenge or delay entry as a first-filer, which is known as “pay-for-delay” settlements.

G Media coverage of AGs

1. <https://www.forbes.com/sites/ikebrannon/2024/01/25/how-drug-companies-use-authorized-generics-to-keep-drug-prices-high/>
2. <https://kffhealthnews.org/news/drugmakers-now-masters-at-rolling-out-their-own-generics-to-stifle-competition/>
3. <https://www.americanprogress.org/article/removing-obstacles-to-generic-drug-competition/>
4. <https://www.precisionvaluehealth.com/2019/04/drug-markers-get-good-pr-via-authorized-generics-but-other-effects-are-less-clear/>
5. <https://www.biosimilardevelopment.com/doc/will-authorized-biologics-disrupt-the-market-for-biosimilars-0001>

H Useful quotes from [FTC \(2011\)](#)

H.1 Quotes on how the brand maker tightly controls the timing of AGs

- (Pg 73)“Early launch, however, “cannibalizes” sales of the brand-name product, starting their decline before ANDA-generic entry has occurred. Brand-name firms have chosen to tightly control the timing of AG launch to avoid unnecessary loss of brand sales and other disadvantages. Launch even a short time before confirmation of ANDA-generic entry is disfavored, and AG supply agreements often include strict penalty provisions or financial terms that enforce launch timing.”
- (Pg 73 footnote) “See CD, Nov. 4, 2004 (“financial risk to brand [of early launch is] significant” because brand erosion cannot be “made up by benefit of 2 week early launch”; that risk is exacerbated if the first generic entrant is delayed for any reason, because the AG would cannibalize the brand during the delay); CD, Feb. 2, 2005 (“Do not put Rx brand at risk (i.e., do not launch in advance of a generic, but can have contracts in place and inventory at wholesalers”)) (emphasis original); CD, Jan. 5, 2007 (“[generic subsidiary] and [brand-name company] agree it is not financially viable to launch an AG prior to launch of other AB rated generics in the US”); CD, Mar. 31, 2004 (disadvantages of launching the AG early include “Uncertainty over ultimate LOE date [that] could cause unnecessary brand erosion; Additional incentive for earlier ‘at risk’ launches by generics; Potential public relations issue; Legislative Risks”).”
- “(Pg 21) The agreements also establish the timing of AG launch. Many agreements provide for launch only after ANDA-generic entry or the date of expected ANDA-generic launch. To ensure that brand sales are not eroded before ANDA-generic competition begins, penalty provisions often apply if the AG marketer launches before generic entry”

H.2 Quotes on outsourcing AG production

- (Pg 77)“Profit-splits, i.e., the percentage of profits that the AG distributor must pay to the brand, are also consistent with the brand’s interest in the revenue stream arising from an AG. Apart from contexts where AG marketing rights derive from patent litigation settlements, marketing agreements generally require AG distributors to pay the brand a large percentage of profits on the AG, typically from 50-92 percent.”
- (Pg 85) “At the same time, some documents also suggest that AG distribution opportunities for generic firms even as early as 2004 were becoming more difficult to obtain, thereby prompting some to re-visit their AG strategies.³⁰ As one generic firm document explains: AGx opportunities and expected returns have both been substantially reduced in recent months, due to the decision of various brand companies to market more of their products through their own AGx subsidiaries, and increased competition among generic companies for available AGx deals...

Revenue and earnings shortfalls by many publicly-traded generic companies due to severe pricing competition has made AGs one of the only ways to maintain growth in the short term; The resulting intense competitive bidding for remaining opportunities has made brand companies realize they are in a position to retain most of the economic value of authorized generic versions of their intellectual property; Historically, generic partners received 20-35% of the economics, but this has now been driven down to 10% or less; The reduced value of AG deals for generic companies will hurt the business models of some companies...”

- “(Pg 23-24) A number of terms affect potential profitability: • Pricing: Neither settlement nor non-settlement agreements restrict the generic company’s authority in setting the price of AGs. Indeed, they frequently affirm that authority. • Transfer Price: The transfer or supply price establishes the amount the generic company pays the brand-name company for the AG. In some agreements, this price is related by express terms to the cost of manufacturing the AG.³⁵ • Profit-Splits: Most agreements provide a profit-split, which governs the distribution of profits between the brand and generic companies from sales of the AG. Many non-settlement agreements require the AG marketer to pay the brand a large percentage of profits on the AG.³⁶

Profit-splits have implications for generic firms' incentives: although AG distribution agreements can give a generic company that was not the first filer the benefit of being able to market during the 180-day exclusivity period, first-filing status provides the possibility of exclusivity unburdened by a profit-split, albeit with the likely expense of patent litigation."

H.3 Quotes on how AGs are usually not aimed at 180-day

- "(Pg 27) Although drug companies marketed many AGs during 180-day exclusivity, they marketed more during non-exclusive periods, and often for drugs that had not been subject to a patent challenge. "
- "(Pg 32) Taken together, the exclusivity and patent certification data in Figures 2-8 and 2-10 indicate that during the period when AGs became common, undermining 180-day exclusivity was unlikely to have been the primary rationale for many AG drugs. This does not preclude a disincentive effect or rationale for those AGs that were marketed during 180-day exclusivity periods, but it suggests that different or additional explanations for AG marketing also should be considered. "
- "(Pg 32) Nonetheless, AGs marketed during exclusivity were in the minority. More AGs were marketed after exclusivity or when no exclusivity had occurred; many of the latter drugs had not been subject to a patent challenge. The rate of AG launch increased as pre-entry brand sales increased, and within most sales levels was similar for ANDA-generic entry with and without exclusivity; there was no consistent pattern in the relative magnitude of rates of AG entry for generic entry with and without exclusivity. These findings suggest that, on average, the anticipated sales level of the brand-name drug has had greater bearing on decisions to launch AGs than whether ANDA-generic entry occurred via exclusivity. The findings, however, do not preclude a disincentive rationale or effect for AGs that were marketed during exclusivity."

H.4 Generics anticipating AG entry

- (Pg v) “Generic companies typically expect authorized generic competition during the 180-day exclusivity period and often build that assumption into their forecasts, which means reducing the expected revenues for the generic challenger. And one company provided an analysis suggesting that this reduction in expected profits had led it to reject or defer filing patent challenges for two products in small markets.”
- (Pg 81-82) “Non-advocacy documents confirm that the introduction of an AG during the exclusivity period substantially erodes the expected profitability of the ANDA. Thus, one internal company document observes that authorized generics are “here to stay,” and “in general cut in half or more the profitability of the value of the 180-day exclusivity.” This estimate is consistent with that in an investment advisory letter, which observed that “[d]ue to market share and pricing erosion at the hands of the authorized player, we estimate that the profits for the ‘pure’ generic during the exclusivity period could be reduced by approximately 60% in a typical scenario.” Another company’s sales forecasts assume an AG, and, because of that assumption, the “[f]orecast is reduced from 100% market share during 6 months of exclusivity to 60% market share.” In fact, various sales and pre-launch forecasts indicate that generic firms routinely assume the presence of an AG, and weave that consideration, along with assumptions about market size, substitution rates, price erosion,¹³ and the likely number of competitors, into their projections of sales and profitability of the ANDA-generic drug during the exclusivity period and beyond. Accordingly, the generic company documents underscore Chapter 3’s quantitative findings of revenue impacts and indicate that, especially for those firms whose business models focus on obtaining exclusivities via Paragraph IV filings, the adverse financial consequences of AGs may be significant.”

I Names of molecule-formulations

Abacavir-oral, Abacavir-lamivudine-zidovudine-oral, Acamprosate-oral, Acarbose-oral, Acetylcysteine-mecobalamin-methylfolate-oral, Acetylsalicylic acid-dipyridamole-oral, Adefovir dipivoxil-oral, Alendronic acid-oral, Alfuzosin-oral, Almotriptan-oral, Alosetron-oral, Amifostine-injectable, Amlodipine-oral, Amlodipine-atorvastatin-oral, Amlodipine-benazepril-oral, Amlodipine-hydrochlorothiazide-valsartan-oral, Amlodipine-telmisartan-oral, Amlodipine-valsartan-oral, Amoxicillin-clarithromycin-lansoprazole-oral, Anastrozole-oral, Apraclonidine-all others, Aripiprazole-oral, Armodafinil-oral, Atorvastatin-oral, Atovaquone-proguanil-oral, Azacitidine-injectable, Azelastine-all others, Aztreonam-injectable, Baclofen-injectable, Balsalazide-oral, Bendroflumethiazide-nadolol-oral, Benztropine-injectable, Benzphetamine-oral, Betamethasone-calcipotriol-all others, Bexarotene-oral, Bicalutamide-oral, Bimatoprost-all others, Bivalirudin-injectable, Budesonide-oral, Buprenorphine-naloxone-oral, C1 inhibitor (human)-injectable, Cabergoline-oral, Calcipotriol-all others, Calcium levofolate-injectable, Candesartan cilexetil-oral, Candesartan cilexetil-hydrochlorothiazide-oral, Capecitabine-oral, Carbidopa-entacapone-levodopa-oral, Carbinoxamine-phenylephrine-oral, Carvedilol-oral, Cefepime-injectable, Cefotetan-injectable, Cefprozil-oral, Ceftriaxone-injectable, Celecoxib-oral, Cevimeline-oral, Chlorcyclizine-codeine-pseudoephedrine-oral, Chlorothiazide-injectable, Choline fenofibrate-oral, Cidofovir-injectable, Cilastatin-imipenem-injectable, Cisatracurium besilate-injectable, Clopidogrel-oral, Colestipol-oral, Cyanocobalamin-methylfolate-pyridoxine-riboflavin-oral, Dactinomycin-injectable, Dantrolene-injectable, Dantrolene-oral, Decitabine-injectable, Desloratadine-oral, Dexmethylphenidate-oral, Dexrazoxane-injectable, Diclofenac-misoprostol-oral, Docetaxel-injectable, Dofetilide-oral, Donepezil-oral, Dorzolamide-all others, Dorzolamide-timolol-all others, Doxercalciferol-injectable, Doxercalciferol-oral, Duloxetine-oral, Dutasteride-oral, Dutasteride-tamsulosin-oral, Enflurane-all others, Enoxaparin sodium-injectable, Entecavir-oral, Epinastine-all others, Epirubicin-injectable, Eplerenone-oral, Epoprostenol-injectable, Eprosartan-oral, Eptifibatide-injectable, Eszopiclone-oral, Etacrynic acid-injectable, Etidronic acid-oral, Exemestane-oral, Famciclovir-oral, Felbamate-oral, Fexofenadine-pseudoephedrine-oral, Finasteride-oral, Fluoxetine-olanzapine-oral, Fluvastatin-oral, Fomepizole-injectable, Fosphenytoin-injectable, Frovatriptan-oral, Galantamine-oral, Ganciclovir-injectable, Gemcitabine-

injectable, Glatiramer acetate-injectable, Glimepiride-oral, Glimepiride-pioglitazone-oral, Granisetron-injectable, Granisetron-oral, Hydrochlorothiazide-irbesartan-oral, Hydrochlorothiazide-telmisartan-oral, Hydrochlorothiazide-valsartan-oral, Hydrocodone-oral, Ibandronic acid-injectable, Ibandronic acid-oral, Ibutilide-injectable, Imatinib-oral, Imiquimod-all others, Irbesartan-oral, Isoniazid-rifampicin-oral, Isradipine-oral, Ivermectin-oral, Lamivudine-oral, Lamivudine-zidovudine-oral, Latanoprost-all others, Leflunomide-oral, Letrozole-oral, Levocetirizine-oral, Levofloxacin-injectable, Levofloxacin-oral, Lincomycin-injectable, Linezolid-oral, Liothyronine-injectable, Liothyronine-oral, Losartan-oral, Mefenamic acid-oral, Melphalan-injectable, Memantine-oral, Mercaptamine-oral, Meropenem-injectable, Metaxalone-oral, Metformin-pioglitazone-oral, Metformin-repaglinide-oral, Methohexital-injectable, Methoxsalen-oral, Methylergometrine-injectable, Methylergometrine-oral, Methylphenobarbital-oral, Miglitol-oral, Mitoxantrone-injectable, Modafinil-oral, Mycophenolate mofetil-oral, Mycophenolic acid-oral, Naftifine-all others, Naratriptan-oral, Nateglinide-oral, Nevirapine-oral, Nicardipine-injectable, Nisoldipine-oral, Octreotide-injectable, Olanzapine-injectable, Olanzapine-oral, Olopatadine-all others, Oxaliplatin-injectable, Oxandrolone-oral, Oxcarbazepine-oral, Paliperidone-oral, Paracetamol-tramadol-oral, Paricalcitol-injectable, Paricalcitol-oral, Pentostatin-injectable, Perindopril-oral, Phenelzine-oral, Phenoxybenzamine-oral, Pimozide-oral, Piperacillin-tazobactam-injectable, Pramipexole-oral, Pravastatin-oral, Prednicarbate-all others, Protriptyline-oral, Quetiapine-oral, Raloxifene-oral, Ramipril-oral, Repaglinide-oral, Rifabutin-oral, Riluzole-oral, Risperidone-oral, Rivastigmine-all others, Rivastigmine-oral, Rizatriptan-oral, Rocuronium bromide-injectable, Ropinirole-oral, Ropivacaine-injectable, Rosuvastatin-oral, Selegiline-all others, Sevoflurane-all others, Sildenafil-oral, Simvastatin-oral, Sirolimus-oral, Spinosad-all others, Stavudine-oral, Sumatriptan-injectable, Sumatriptan-oral, Tacrolimus-all others, Tacrolimus-oral, Tamsulosin-oral, Tazarotene-all others, Telmisartan-oral, Temozolomide-oral, Teniposide-injectable, Tetrabenazine-oral, Tetracosactide-injectable, Tiagabine-oral, Tinidazole-oral, Tolcapone-oral, Topiramate-oral, Topotecan-injectable, Trandolapril-oral, Trandolapril-verapamil-oral, Tranexamic acid-injectable, Tranylcypromine-oral, Travoprost-all others, Trimipramine-oral, Trosipium-oral, Valaciclovir-oral, Valganciclovir-oral, Valproate semisodium-oral, Valsartan-oral, Voriconazole-injectable, Voriconazole-oral, Zafirlukast-oral, Zidovudine-oral, Ziprasidone-oral, Zoledronic acid-injectable, Zolmitriptan-oral, Zolpidem-oral, Zonisamide-oral

J Descriptive work with filing data

The data for this section comes from [Feldman et al. \(2016\)](#), which is publicly available in PDF form. We thank Robin Feldman for sharing the spreadsheet version of the data with us.

Table 17: Variation in approval dates across molecule-formulations.

	25%	30%	40%	50%	75%	90%
Approval Month Range	0.0	0.0	5.2	21.0	49.5	83.2
Approval Month IQR	0.0	0.0	0.85	7.5	22.25	43.1
Approval Month P90 minus P10	0.0	0.0	4.16	18.4	39.0	61.08

Table 17 shows the variation in approval times across molecule-formulations. For each molecule-formulation, we calculate the range/IQR/P90-P10 of approval times across all ANDAs. The table then shows the distribution of these measures across molecule-formulations. For example, the median molecule-formulation has an approval month range of 21 months, an IQR of 7.5 months, and a P90-P10 of 18.4 months.

Table 18: Variation in filing dates across molecule-formulations.

	10%	25%	50%	75%	90%
Filing Month Range	2.2	13.5	40.0	67.5	96.4
Filing Month IQR	0.95	5.25	16.0	28.375	43.2
Filing Month P90 minus P10	1.76	9.3	31.0	50.0	72.48

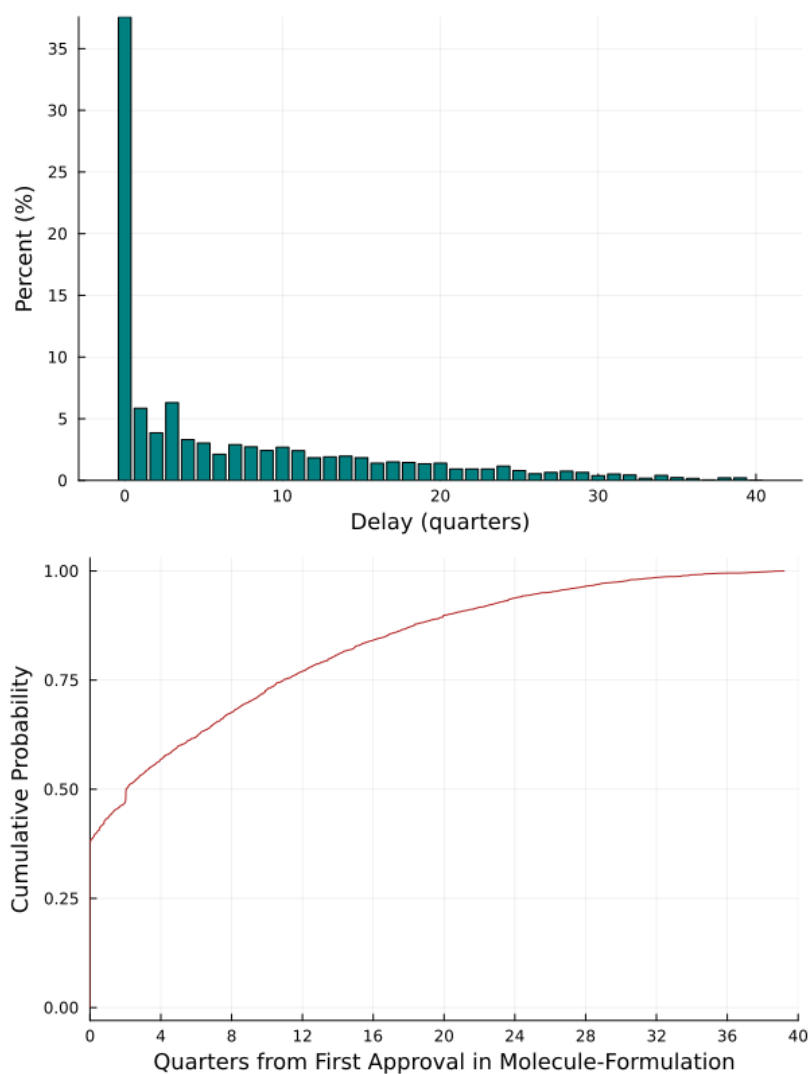
Table 18 shows the variation in filing times across molecule-formulations. For each molecule-formulation, we calculate the range/IQR/P90-P10 of filing times across all ANDAs. The table then shows the distribution of these measures across molecule-formulations. For example, the median molecule-formulation has a filing month range of 40 months, an IQR of 16 months, and a P90-P10 of 31 months.

Table 19 shows the distribution of months to approval across all ANDAs. The median ANDA takes 35.3 months to approval, while the 10th and 90th percentiles are 17 and 70.6 months, respectively.

Table 19: Distribution of months-to-approval across all ANDAs.

	10%	25%	50%	75%	90%
Months to Approval	17.0067	22.8167	35.3	50.0833	70.5867

Figure 7: Histogram of delay between first entrant and later entrants in a molecule-formulation.



In Figures 7, we recreate the “entry date relative to LOE” graphs in the main text using data on approval dates.

To do so, for each molecule-formulation, we find the first entrant (i.e. the ANDA with the earliest approval date). We then calculate the delay between the approval date of each later entrant and that of the first entrant. The top graph in Figure 7 shows a histogram of these delays across all molecule-formulations, while the bottom graph shows a CDF of these delays across all molecule-formulations.

The results are quite similar to our findings using launch dates from the IQVIA dataset.